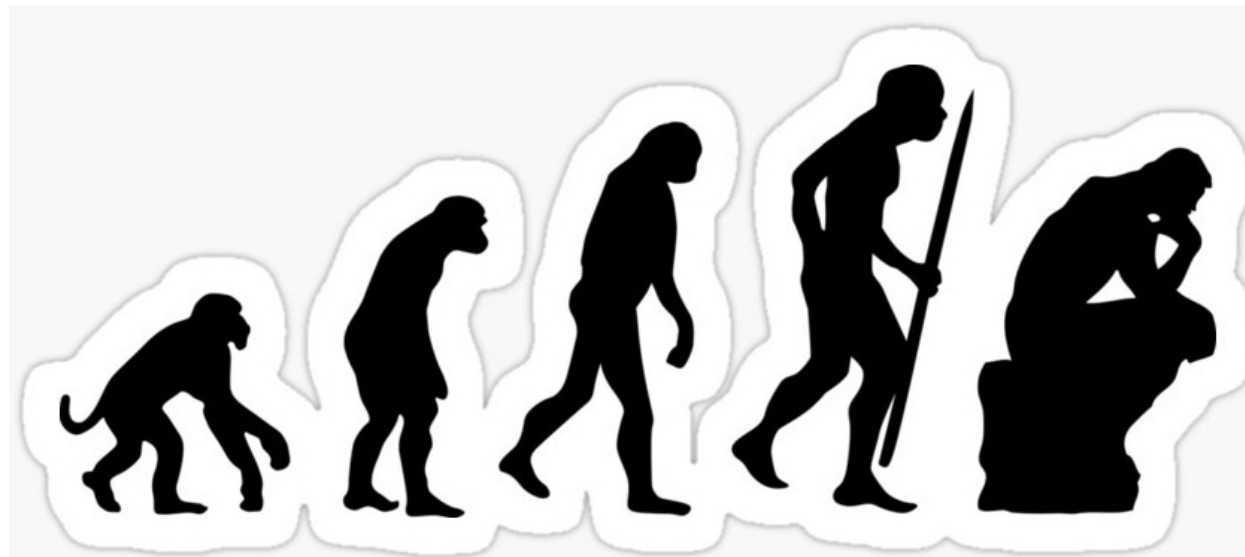


The theory of evolution as the grand story of biology

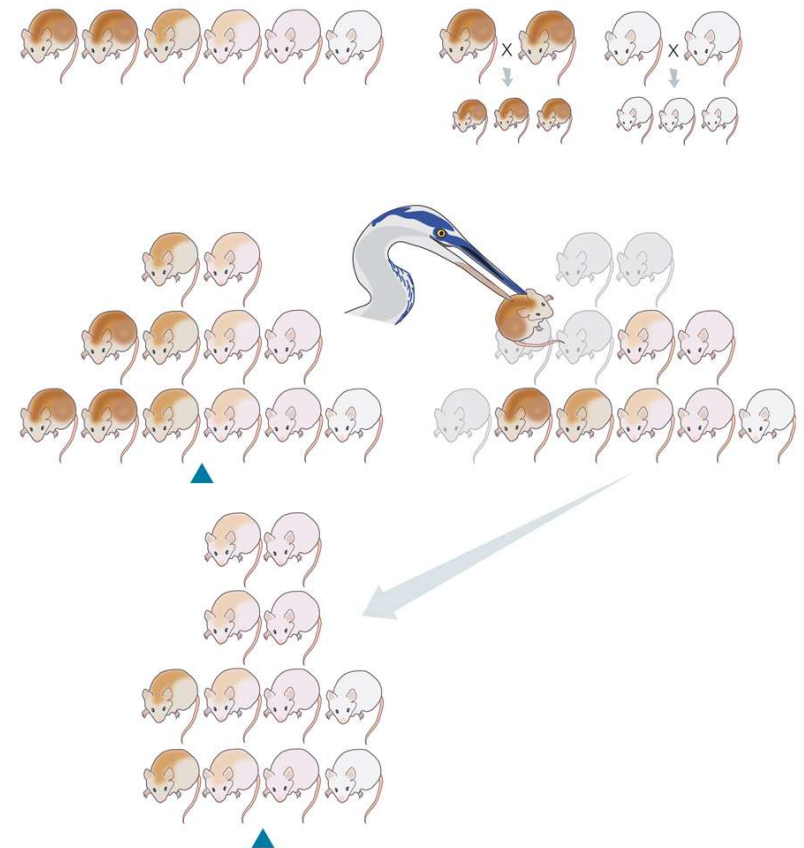
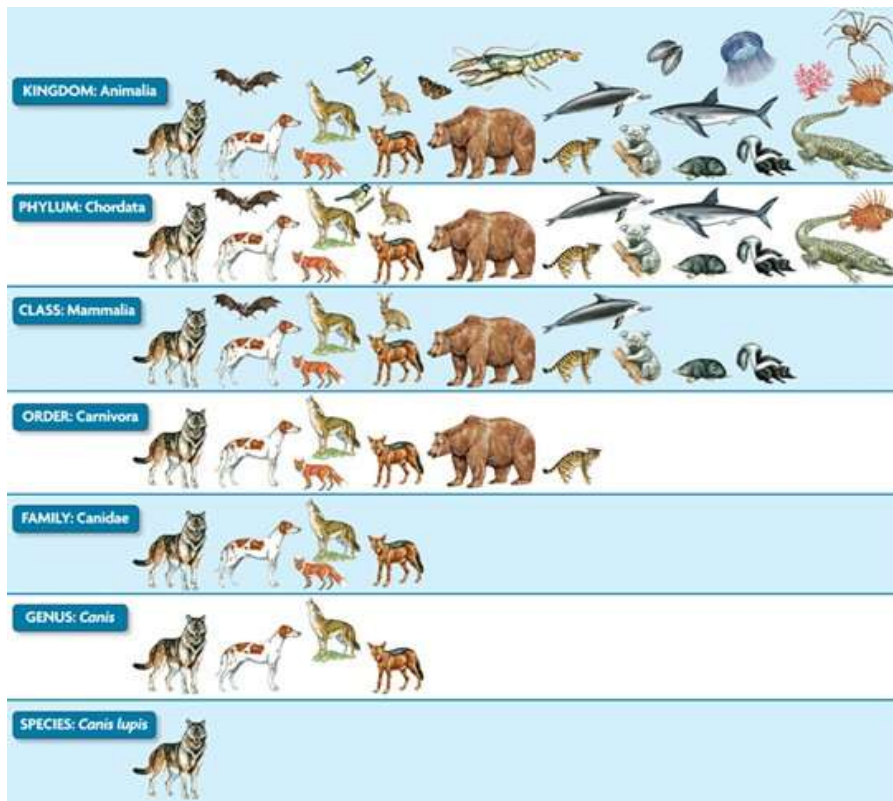
But not as religion or myth

Duur Aanen

Laboratory of Genetics
Wageningen University
The Netherlands



Evolution is a scientific theory that explains the patterns of taxonomy and the process of adaptation



Evolutionary biology is a “regular” branch of the natural sciences

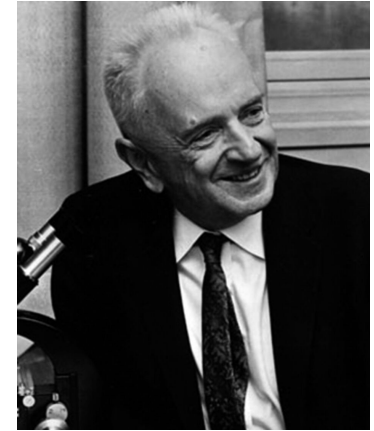
- Develops hypotheses and tests those with observations and experiments
- Modern theory of evolution has a solid base but is an active field of research and is not “finished” :
 - What is the role of chance versus natural selection?
 - How can reproductive isolation arise?
 - How can cooperation evolve?
 - ...
 - ...

But evolutionary biology seems special

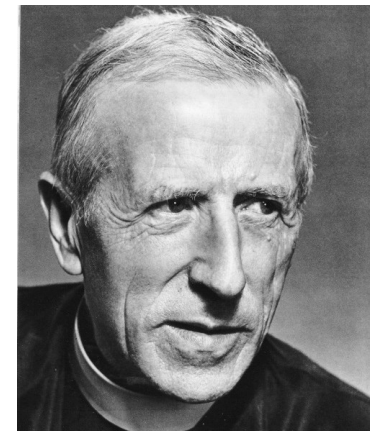
The fundament of biology

“Nothing in biology makes sense, except in the light of evolution”

“Evolution is a light which illuminates all facts, a trajectory which all lines of thought must follow – this is what evolution is.”

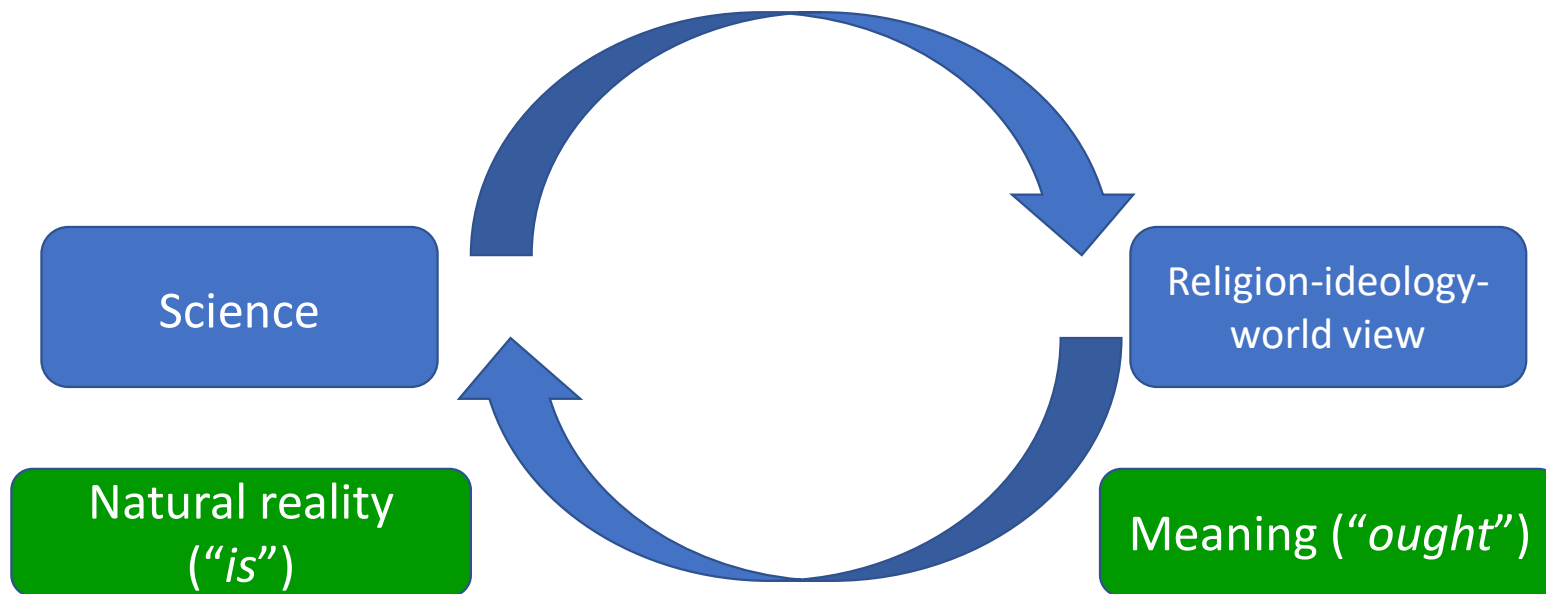


Theodosius Dobzhansky
(1900-1975)



Pierre Teilhard de Chardin
(1881 –1955)

Is evolution móre than science?



'I believe in Darwin. And this means that you continuously have to adapt to remain successful'

“The environment changes continuously, so you have to invent yourself over and over again and to adapt. I have a firm belief in that.”

“Not only for species, also for organisations, countries, everyone personally, everything. That has always inspired me, also when I started to change and adapt DSM. Darwin is an inspiration for me.”



Feike Sijbesma. Former CEO of DSM (*Trouw*, 25 February 2023)

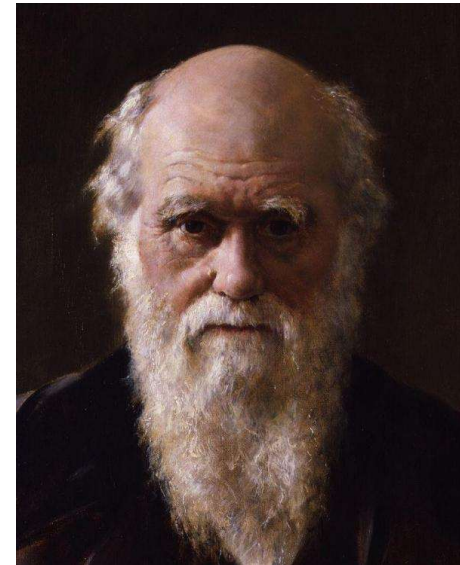
Is evolution more than science?

- The history of Darwinism
- What is evolutionary biology and what is it not?
- What makes evolution difficult to accept for some people?

Is evolution more than science?

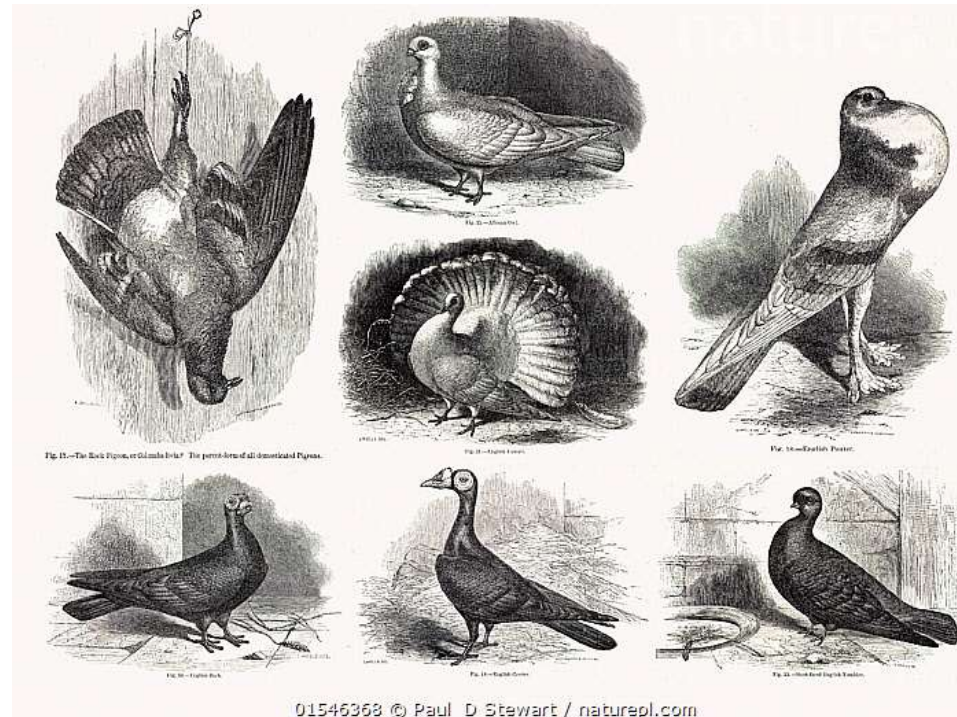
- The history of Darwinism
- What is evolutionary biology and what is it not?
- What makes evolution difficult to accept for some people?

The history of Darwinism: 1859



Charles Darwin
(1809-1882)

ON THE ORIGIN OF SPECIES — CHARLES DARWIN



01546368 © Paul D Stewart / naturepl.com

Darwin: life evolved via *Descent with modification*

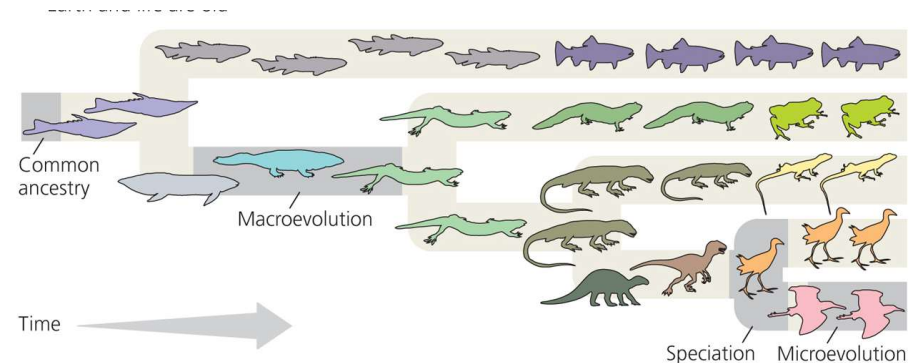
1. Species can change

2. Populations can split and become different (speciation)

3. New living form descend from old.

4. All life is related via common ancestors

5. The earth and life are old



The Wilberforce-Huxley debate in Oxford in 1860

The classical story: a clash between Science and Religion. Wilberforce tried to ridicule Huxley (*"Darwin's bulldog"*) by asking:

"Is it on your grandfather's or grandmother's side that you claim descent from the apes?"

Whereupon Huxley retorted:

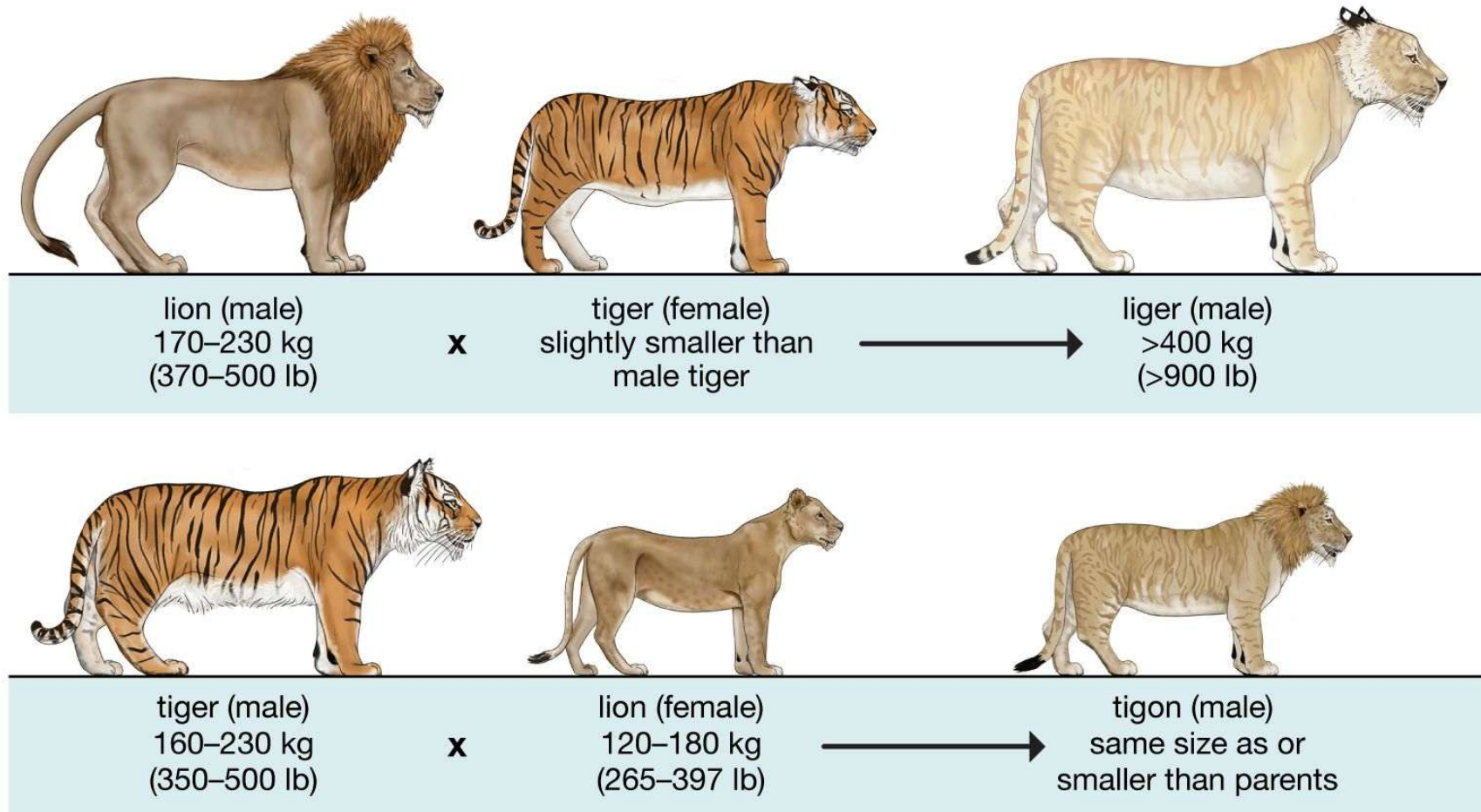
"I would rather be descended from an ape than a bishop."



In reality, Wilberforce had *scientific* criticism on Darwin's theory

- Over the course of human history, there was no record of new species.
- Selective pressures, which he admitted certainly were there, did not cause a new *species*
- The sterility of hybrids told strongly in favour of the fixity of species.

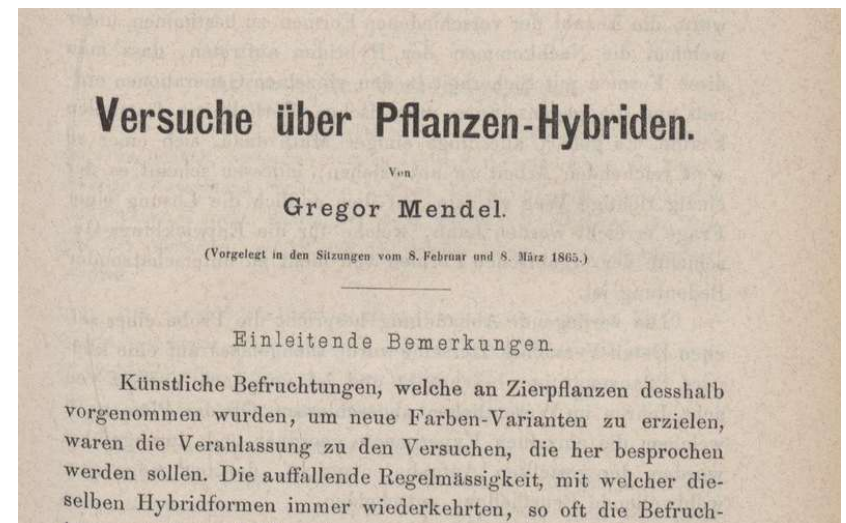
Examples of hybrid mammals: the liger and the tigon



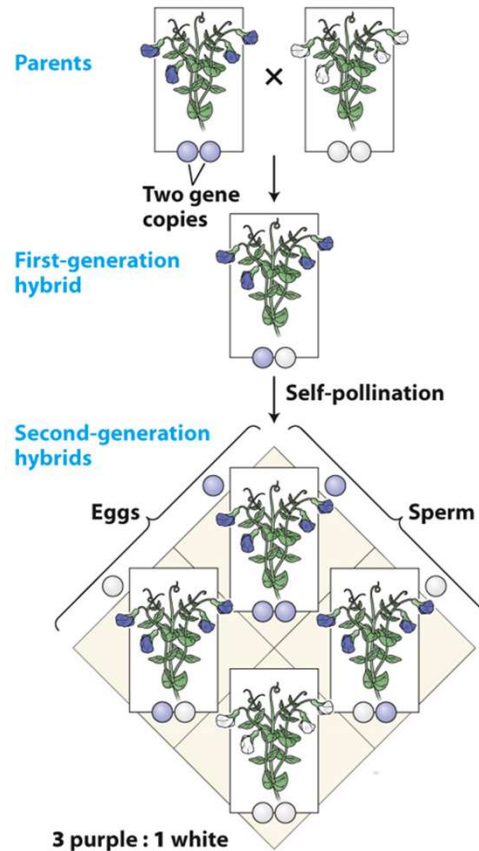
3ft 
1m 

There were additional *scientific* arguments against Darwin's theory at that time

Most importantly, Darwin did not have a good understanding of genetics.



1865 Mendel presented his laws of inheritance



Pure lines
different alleles
gametes have 1 copy

Uniform F1
dominance / recessiveness

Segregating F2 (3:1)

Mendel concluded (1) that genes behave like particles and do not blend together, and (2) one allele is dominant to the other.

Mendel published his work in 1865, but his work remained undiscovered until early 1900's

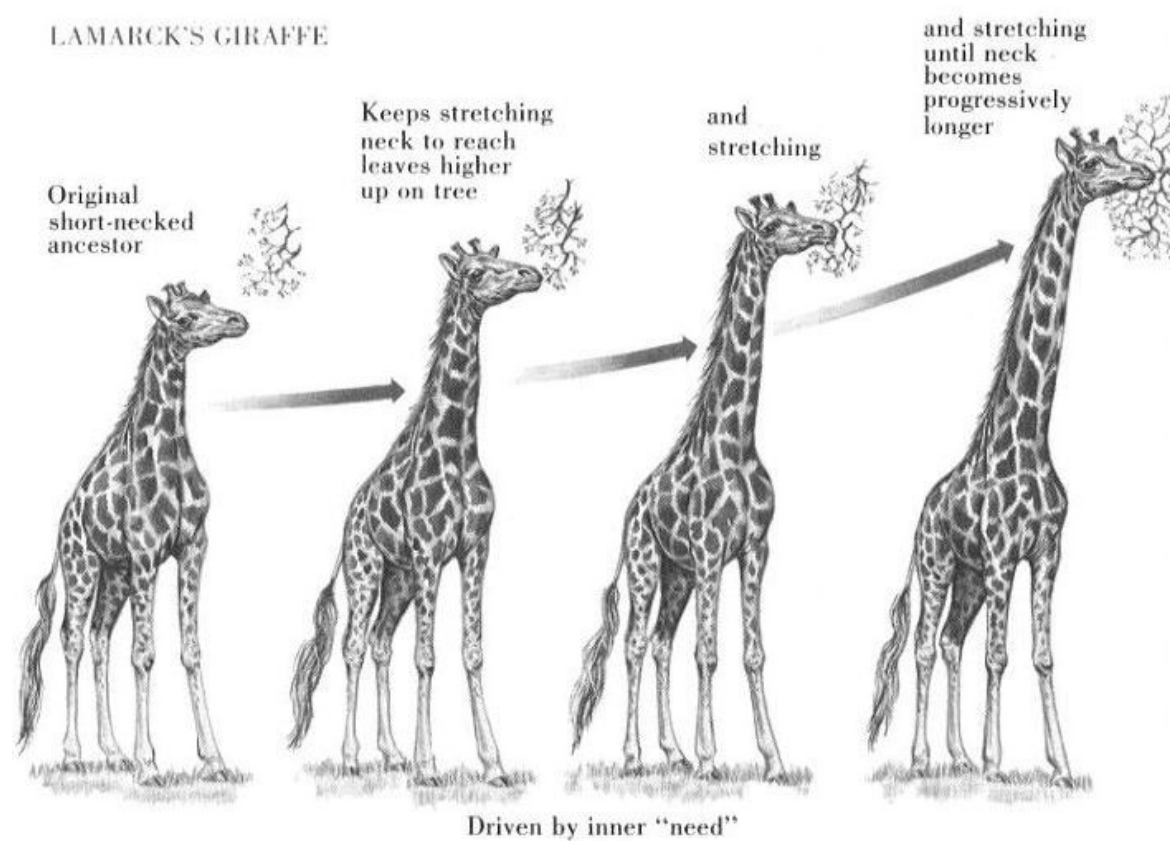
And all kinds of other theories were around...

Some history...

Jean-Baptiste de Lamarck (1744-1829): mutations are acquired. Environment influences genetic constitution:
'Experience' is heritable.

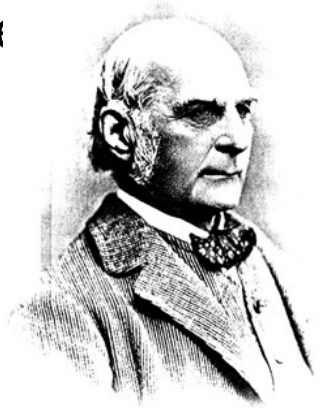


‘Experience by an individual is inherited by the offspring’



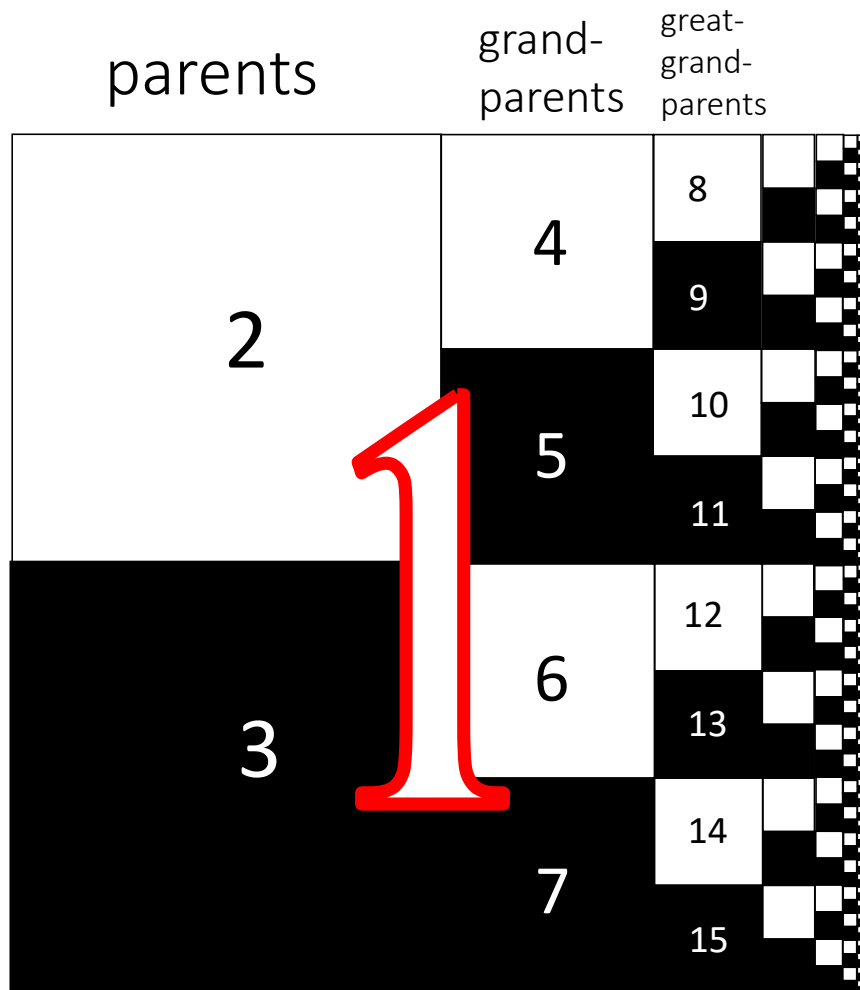
Some history...

- Lamarck (1744-1829): mutations are acquired
- Francis Galton (1822-1911): a quantitative model of inheritance and acquisition of new 'mutations'





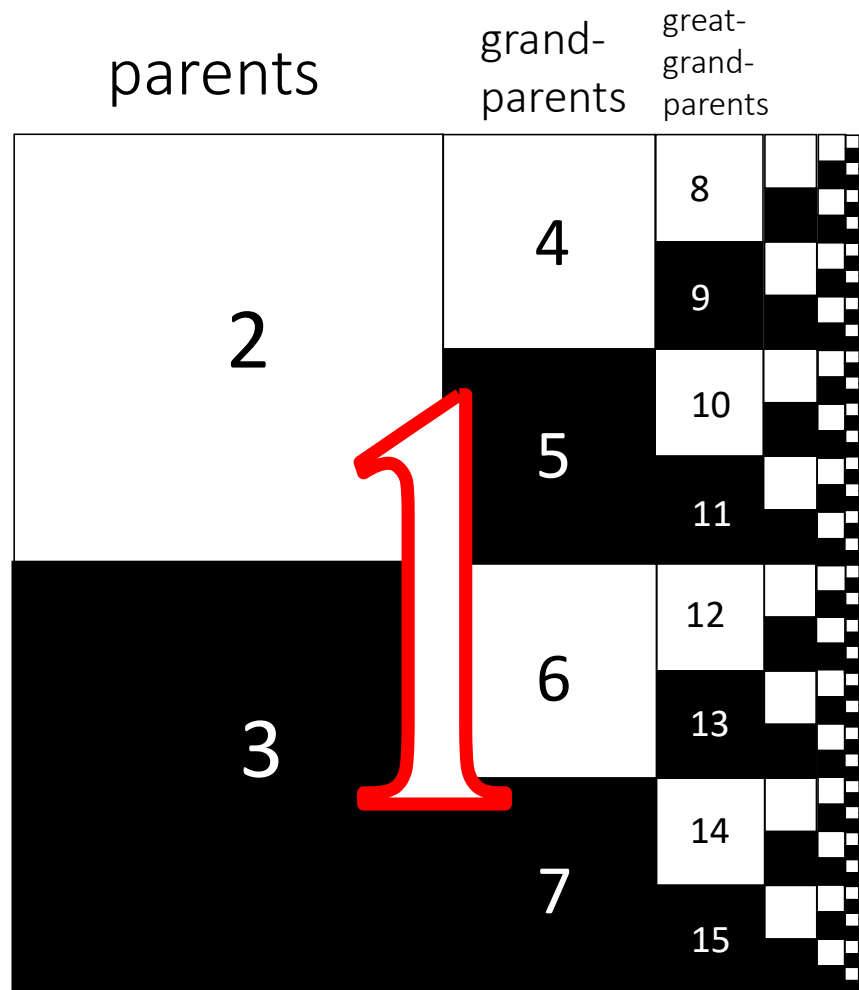
Francis Galton
(1822-1911)



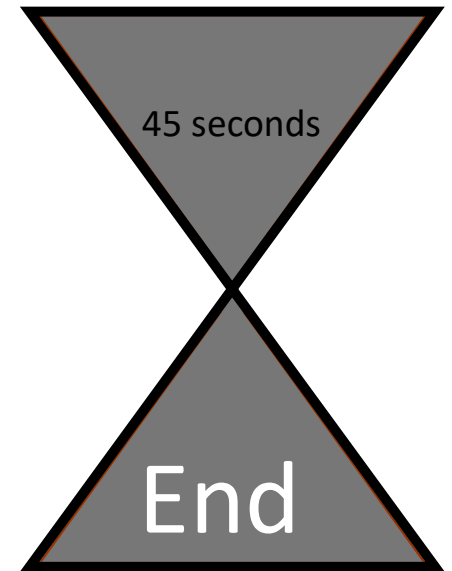
Francis Galton, A diagram of heredity,
Nature, 1898



Francis Galton
(1822-1911)



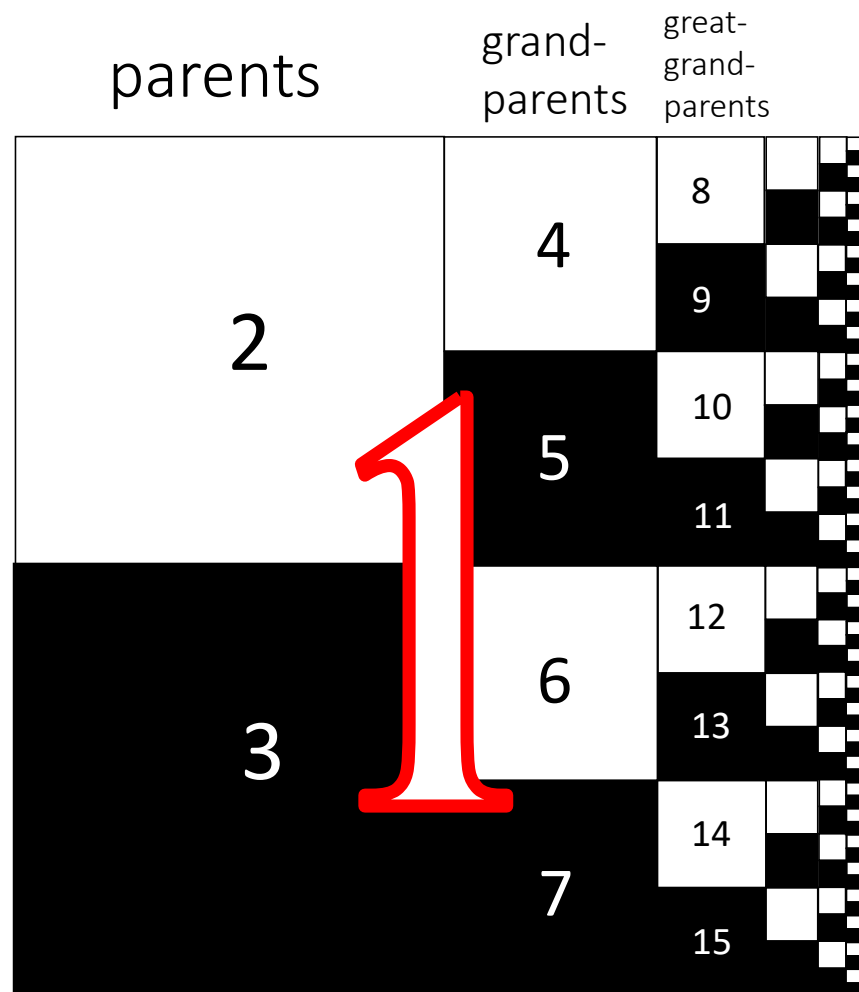
Francis Galton, A diagram of heredity,
Nature, 1898



Individual 1 inherits from her ancestors...



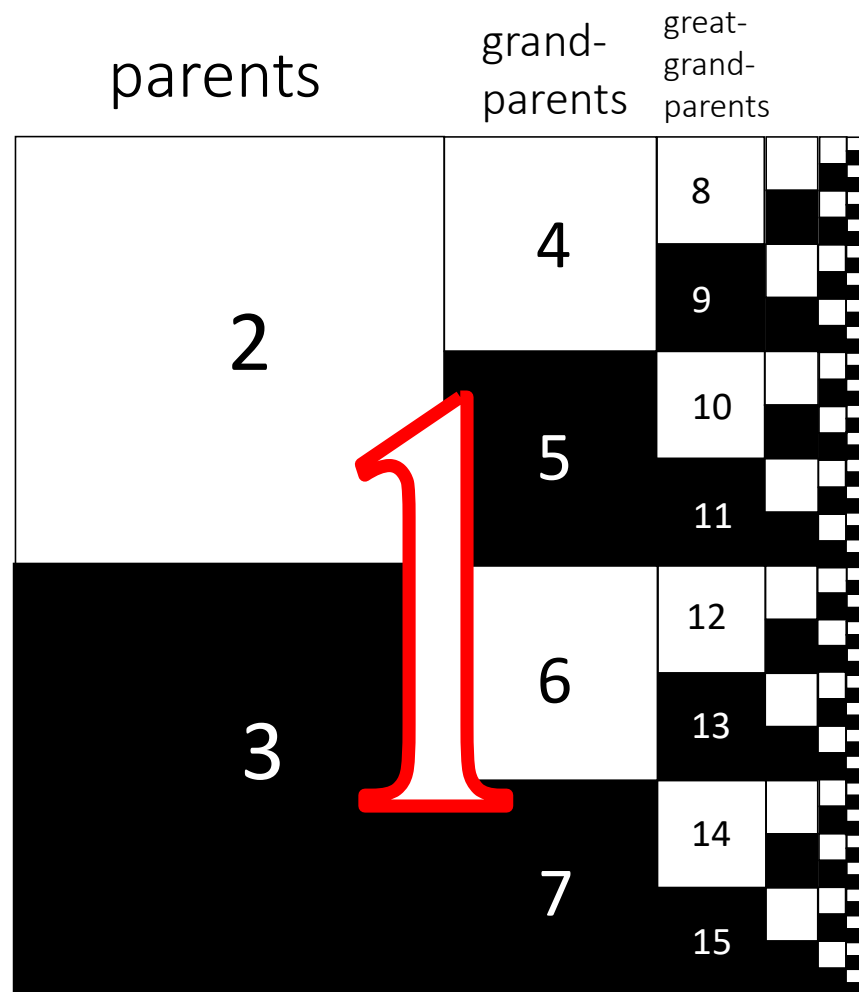
Francis Galton
(1822-1911)



Francis Galton, A diagram of heredity,
Nature, 1898



Francis Galton
(1822-1911)

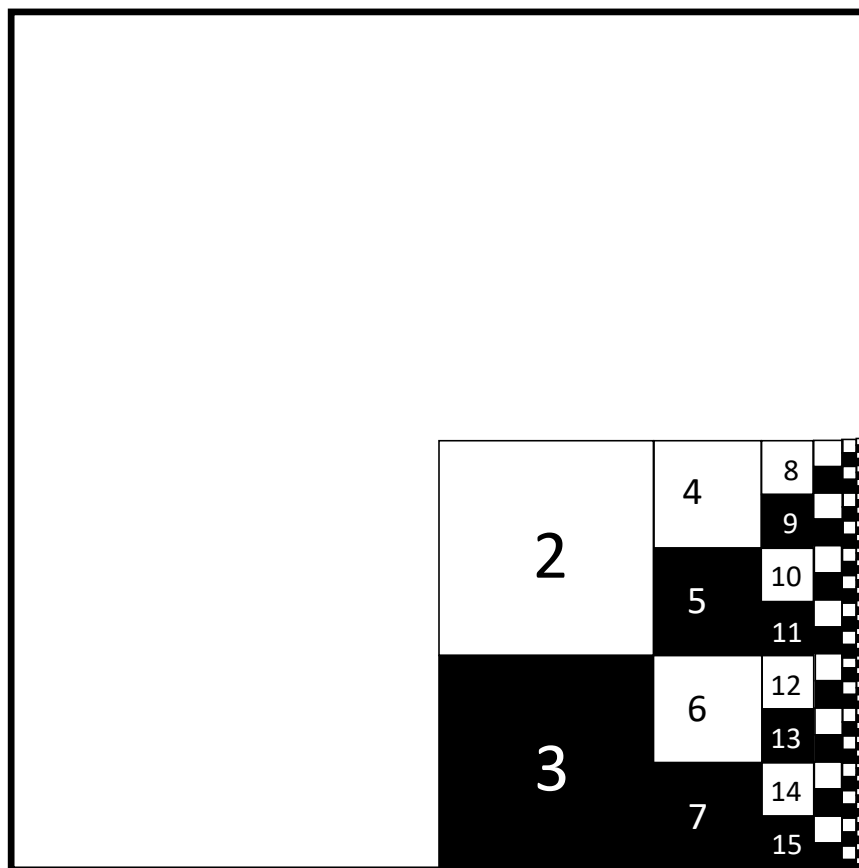


Francis Galton, A diagram of heredity,
Nature, 1898



Francis Galton
(1822-1911)

1 's contribution to a child...

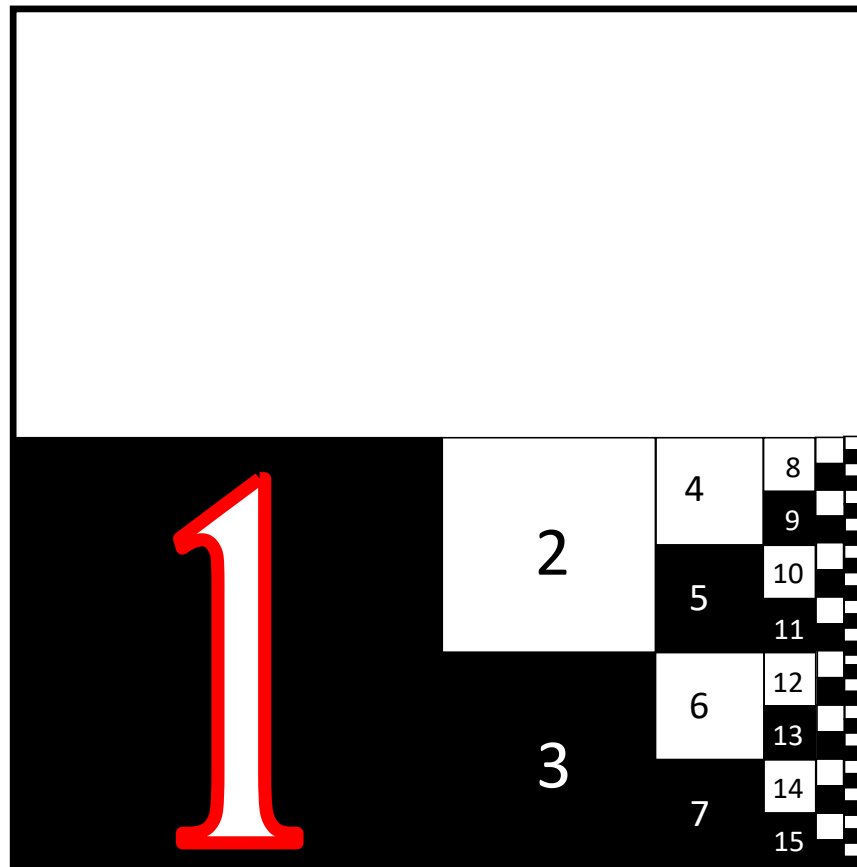


What she inherited herself...



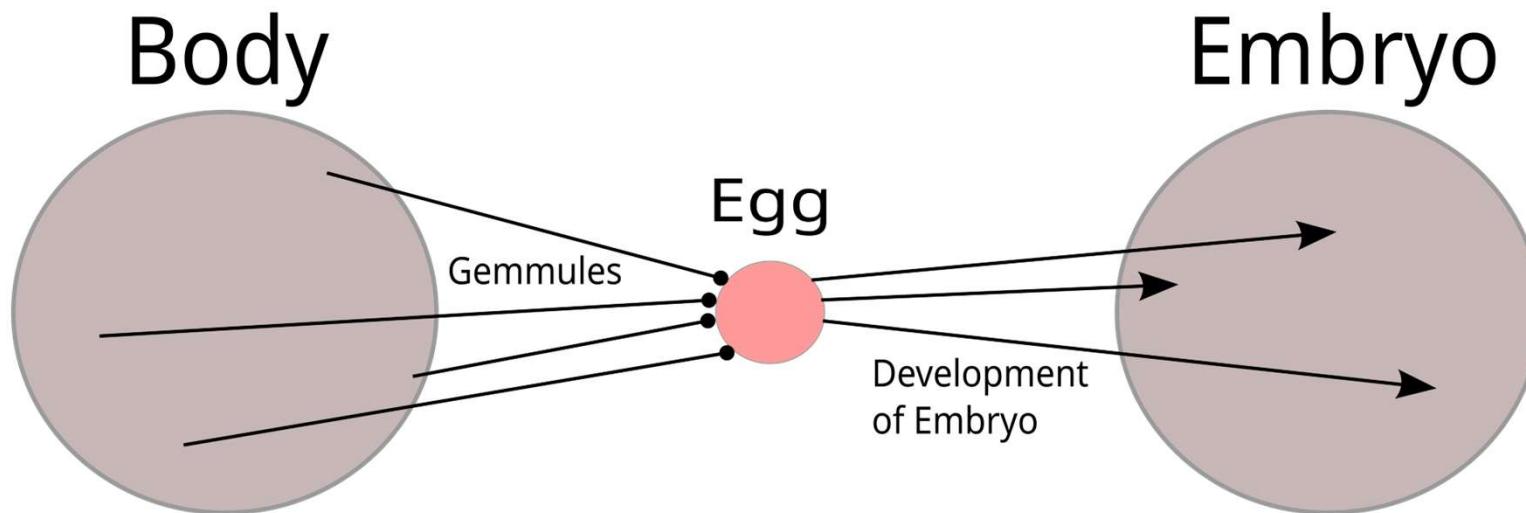
Francis Galton
(1822-1911)

1 's contribution to a child...

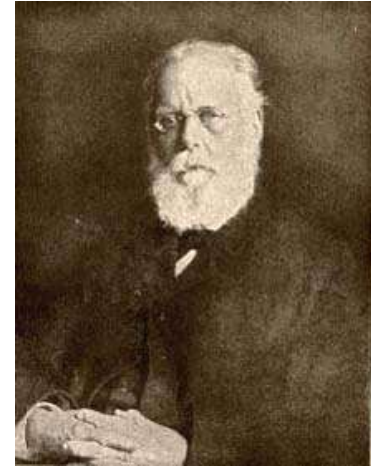
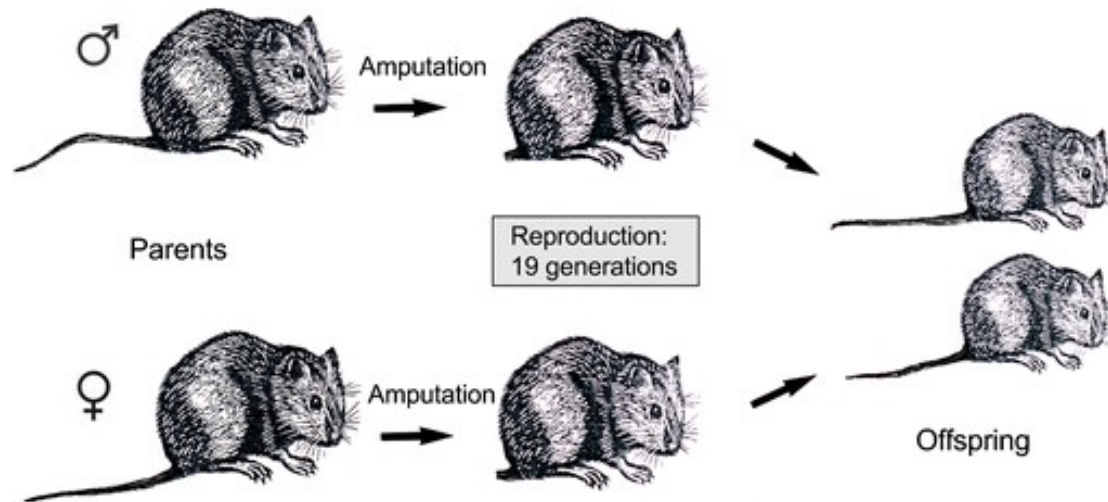


+what she has 'experienced' What she inherited herself...
or 'collected' during her life!

Darwin had a variant of acquired inheritance, the theory of 'blending inheritance'



Weismann tested theory of acquired inheritance experimentally (1890)



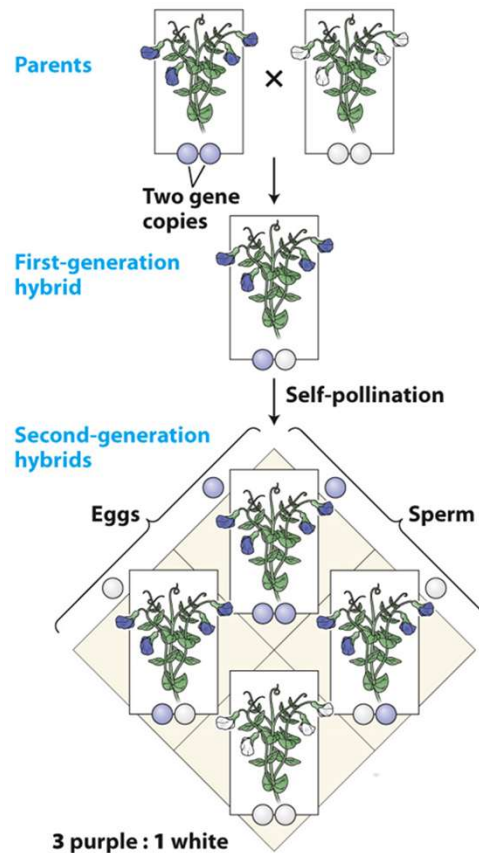
August Weismann
(1834-1914)

“901 young were produced by 19 generations of artificially mutilated parents, and yet there was not a single example of a rudimentary tail or of any other abnormality in this organ.”

**ESSAYS UPON HEREDITY AND KINDRED BIOLOGICAL
PROBLEMS, VOLS 1 AND 2**
AUGUST WEISMANN

So, theories of acquired inheritance and blending inheritance were *wrong*! We do not acquire new genetic information from experience...

1865 Mendel presented his laws of inheritance



Pure lines
different alleles
gametes have 1 copy

Uniform F1
dominance / recessiveness

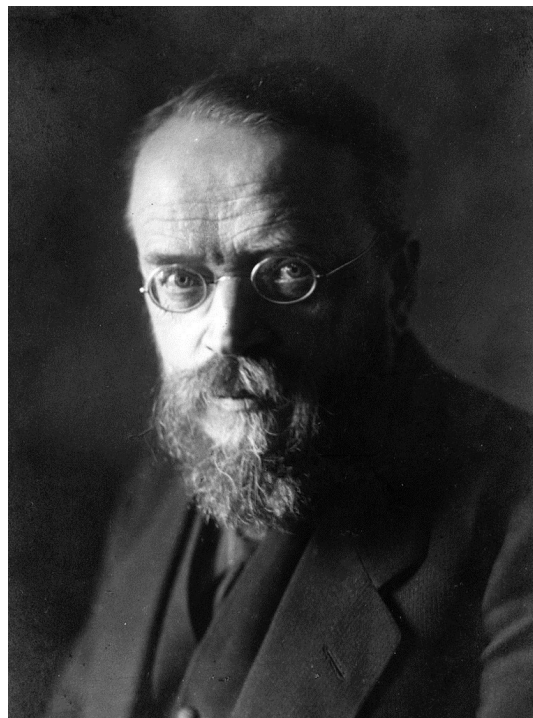
Segregating F2 (3:1)

Mendel concluded (1) that genes behave like particles and do not blend together, and (2) one allele is dominant to the other.

But it lasted until 1900 before Mendel's laws were 'rediscovered'...



Hugo de Vries

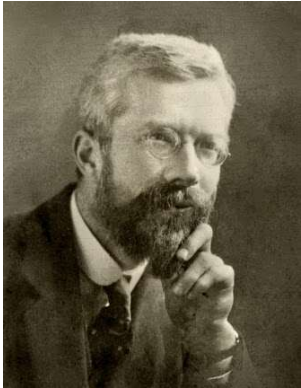


Carl Correns

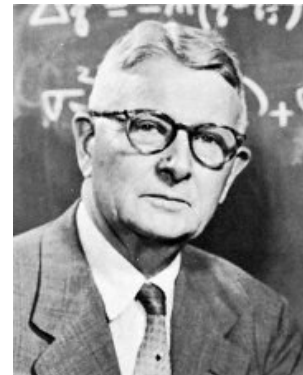
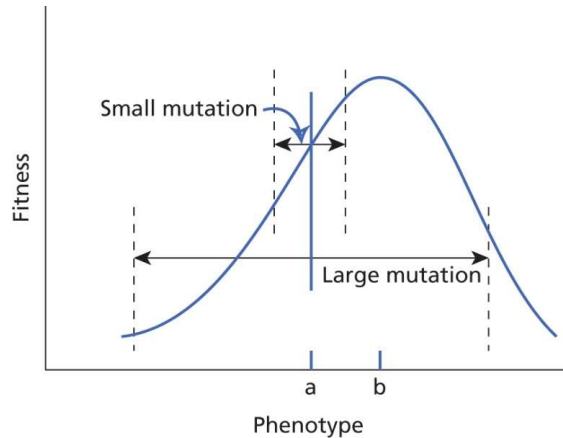


Erich von Tschermak

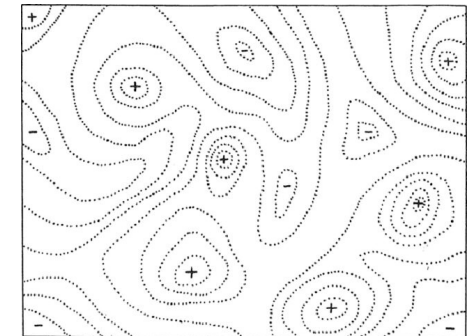
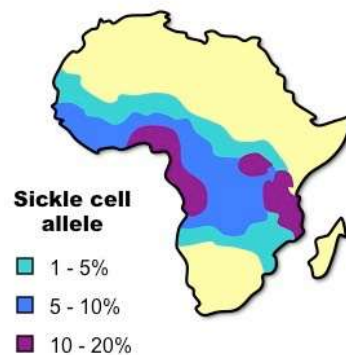
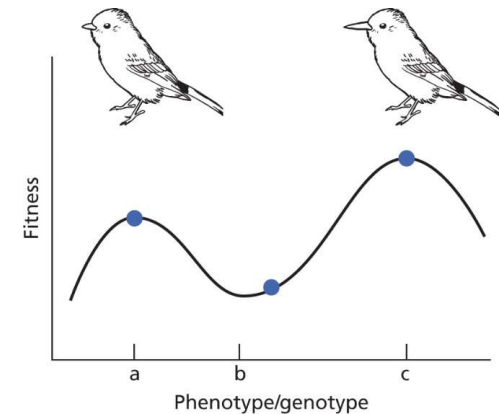
Fisher, Wright & Haldane integrated Mendel with evolution → The Neo-Darwinian synthesis



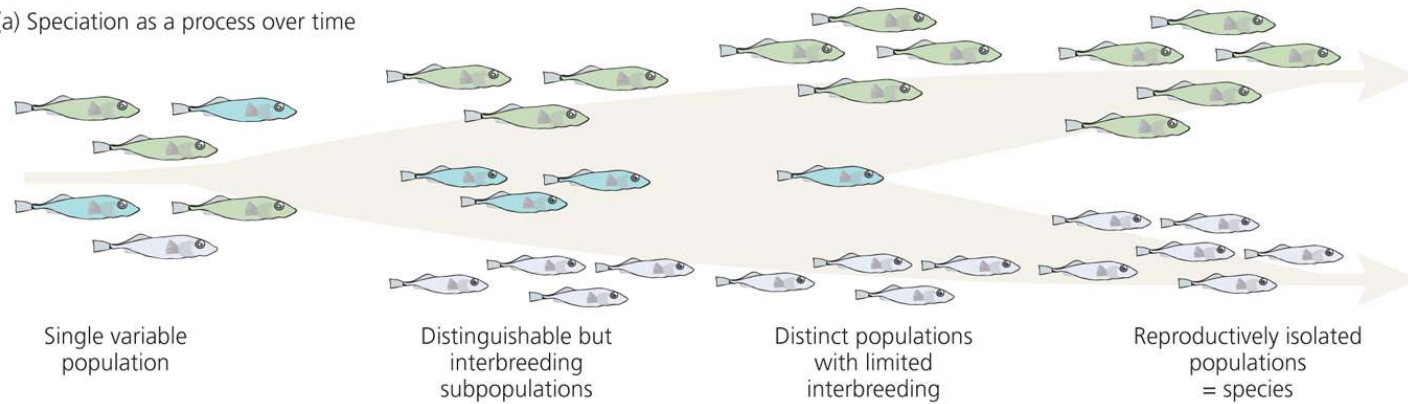
Ronald Fisher



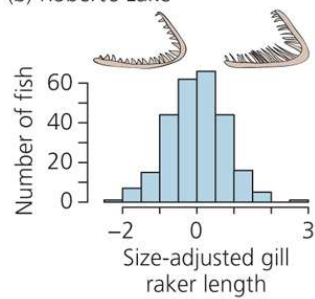
Sewall Wright



(a) Speciation as a process over time



(b) Robert's Lake



(c) Robert's Lake and Outlet
Robert's Creek



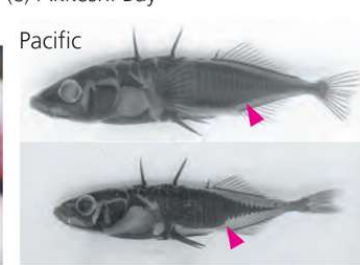
Robert's Lake

(d) Paxton Lake
Benthic (lake bottom)



Limnetic (open water)

(e) Akkeshi Bay

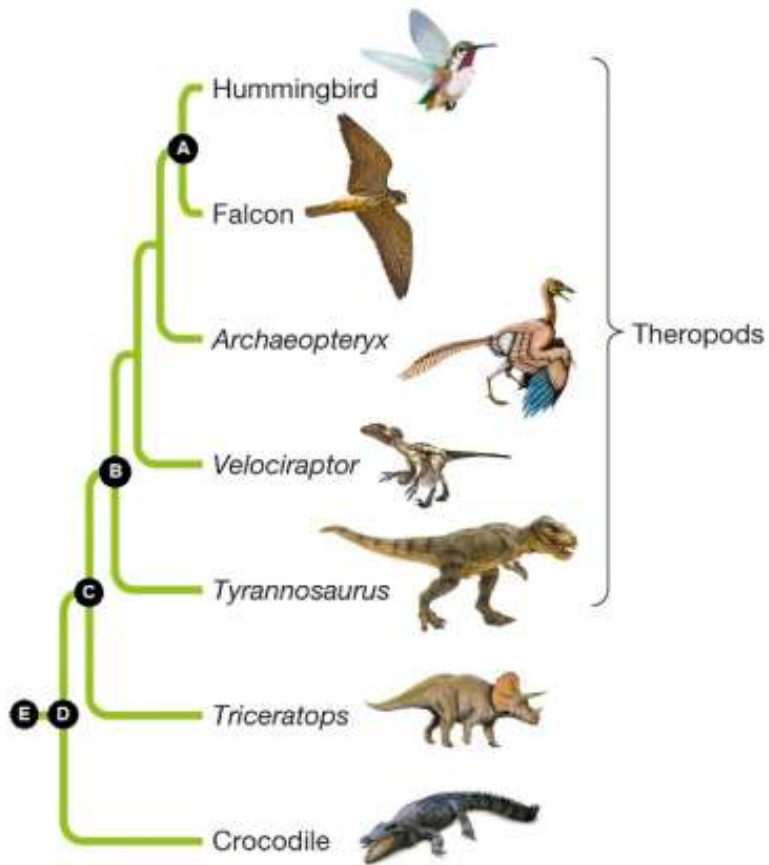


Pacific

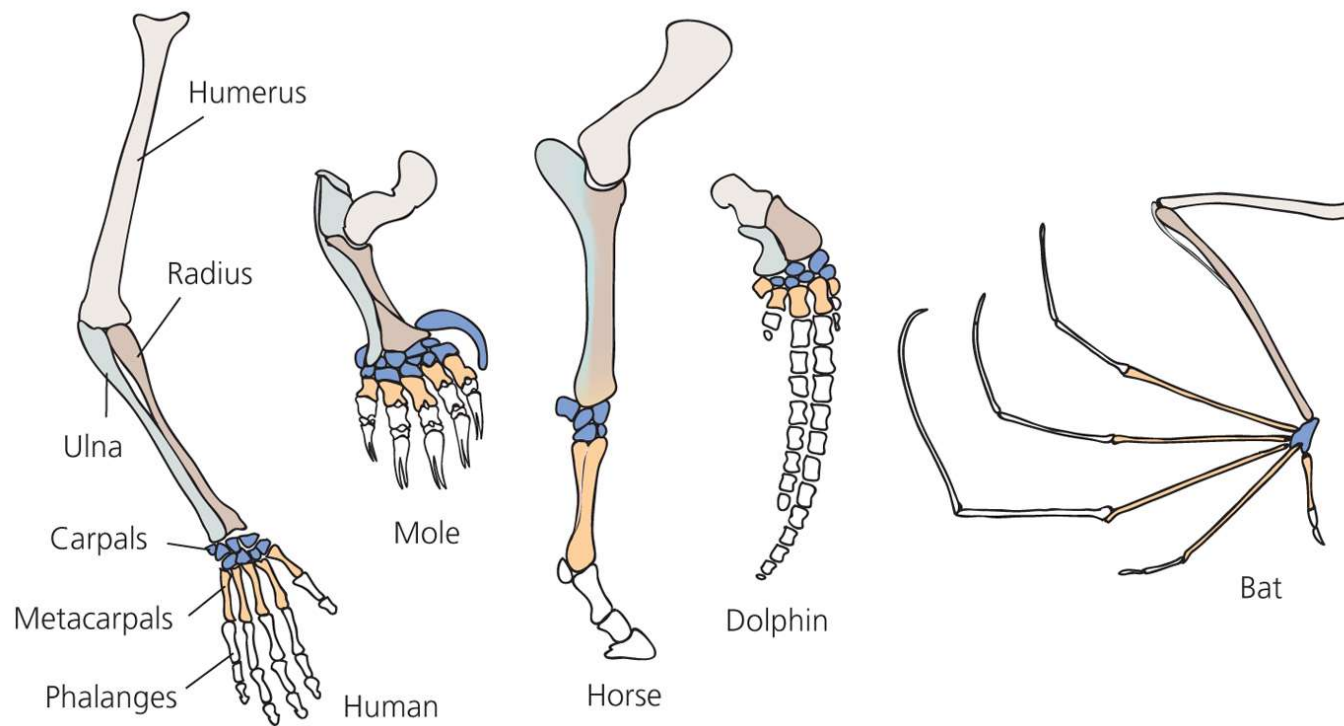
Japan Sea



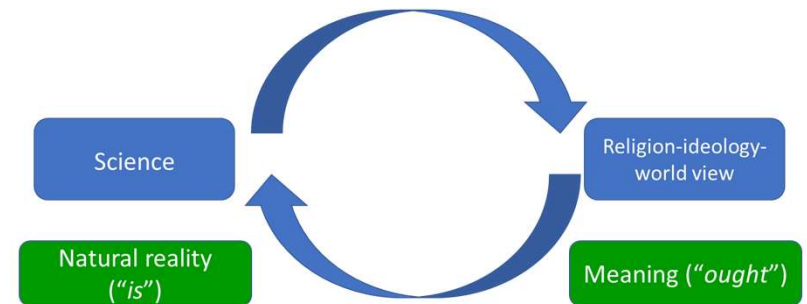
“Missing links” have been found



Common descent is apparent from homologous structures



Conclusions on the history of Evolution



It is often argued that Darwin's theory provided an inspiration for some world views, such as capitalism ("survival of the fittest" can be applied to society; **social Darwinism**) and conservatism ("gradual change is to be preferred over revolutions").

However, it seems fair to say that the reverse was as true. So changing world views were an important inspiration for Darwin.

So, Darwin's idea initially was more based on an intuition than on scientific arguments. Later developments showed that Darwin was right.

Is evolution more than science?

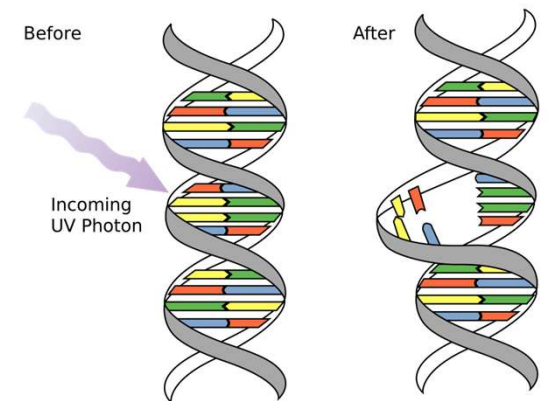
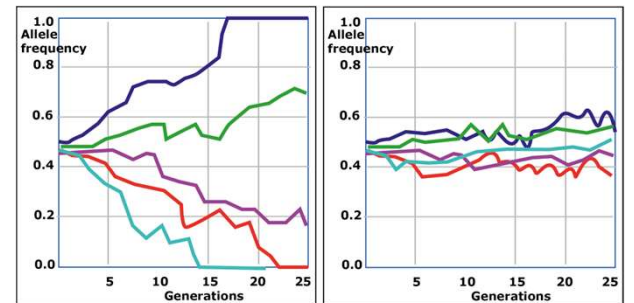
- The history of Darwinism
- What is evolutionary biology and what is it not?
- What makes evolution difficult to accept for some people?

Is evolution more than science?

- The history of Darwinism
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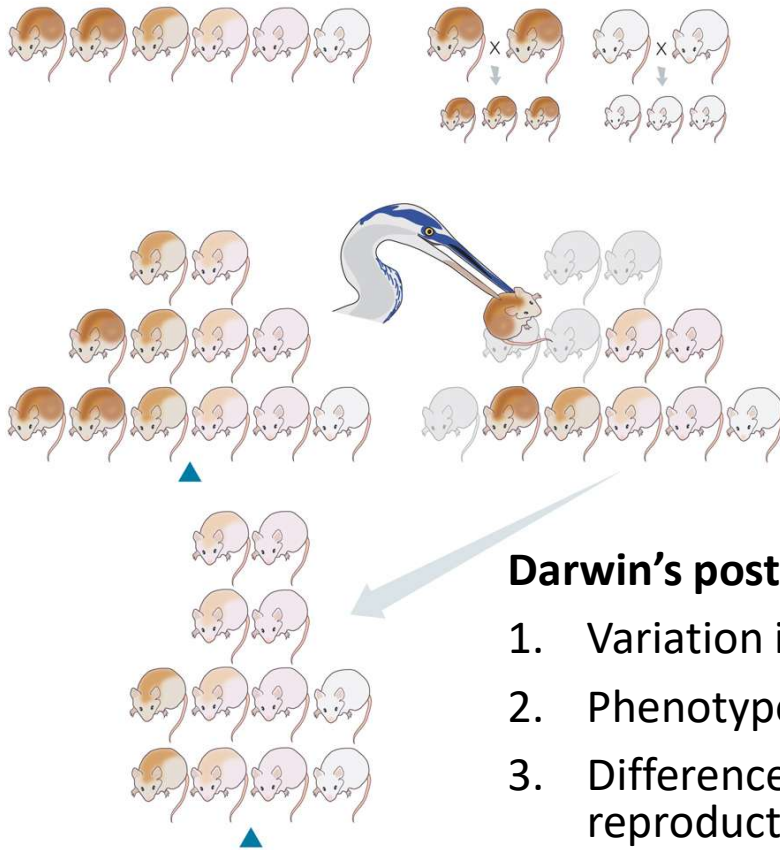
'Chance' plays an important role in evolution

- Patterns: Meteorite causing extinction of large land animals created an opportunity for placental mammals;
- Genetic drift: chance effects in small populations
- Random mating: Hardy Weinberg
- Mutations are undirected (*with a few exceptions*). This is an empirical finding, in theory mutation could be a directed process



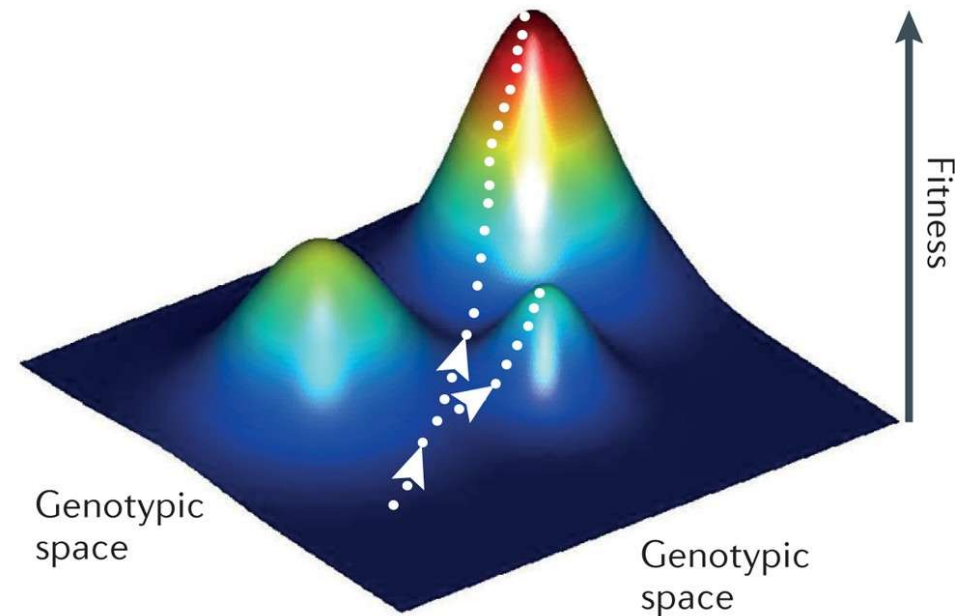
Natural selection gives 'direction'...

... towards higher fitness



Darwin's postulates:

1. Variation in phenotype
2. Phenotype is heritable
3. Differences in survival and/or reproduction (fitness)
4. Correlation between phenotype and fitness

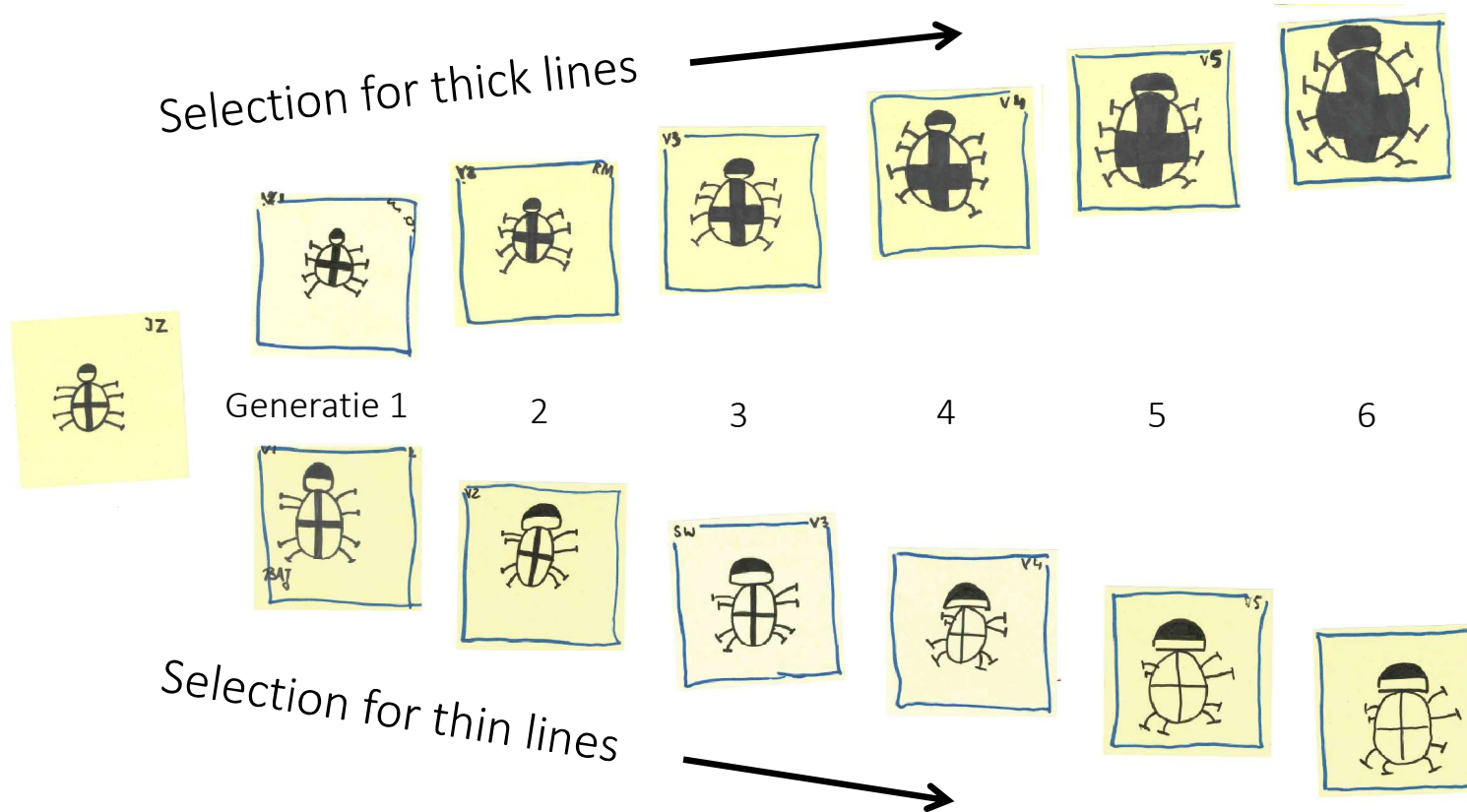


De Visser and Elena, 2014

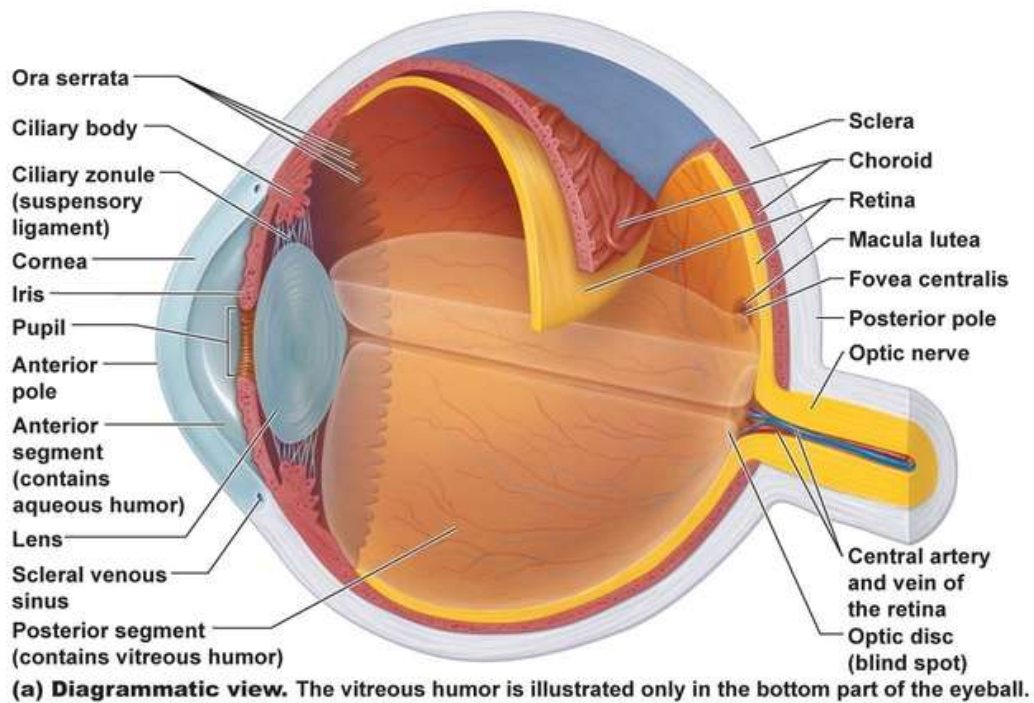
The concept of natural selection more generally applicable



Jelle Zandveld



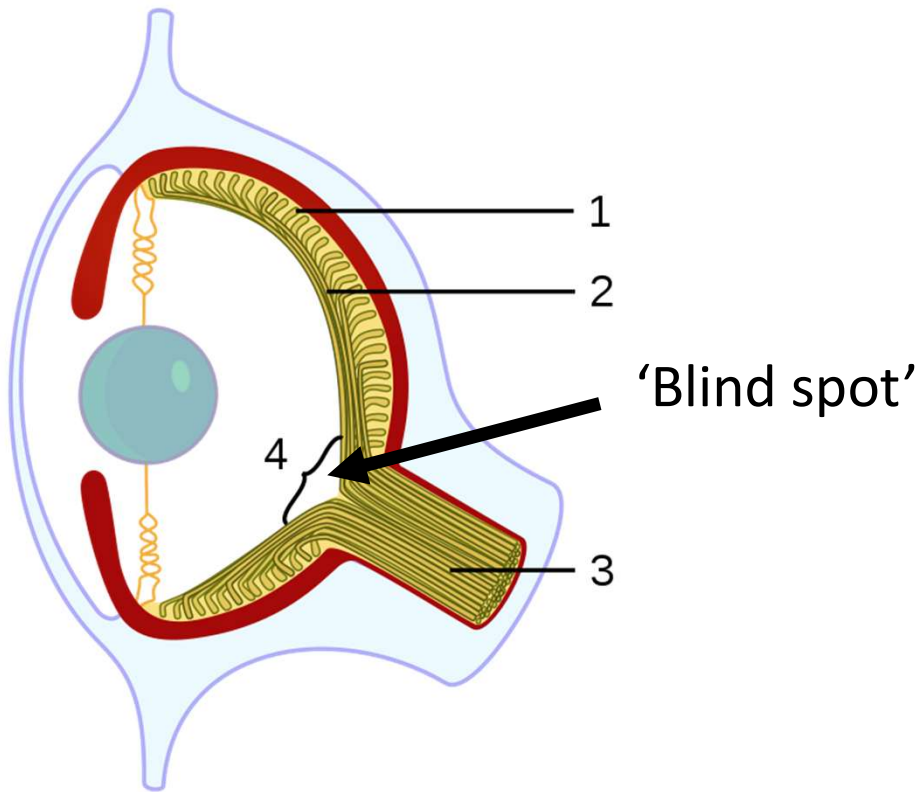
Natural selection 'shapes adaptation' ('adaptive design')



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‘Adaptive’, yes, but ‘optimal design’?



Pattern of change if octopus eye and vertebrate eye are convergent

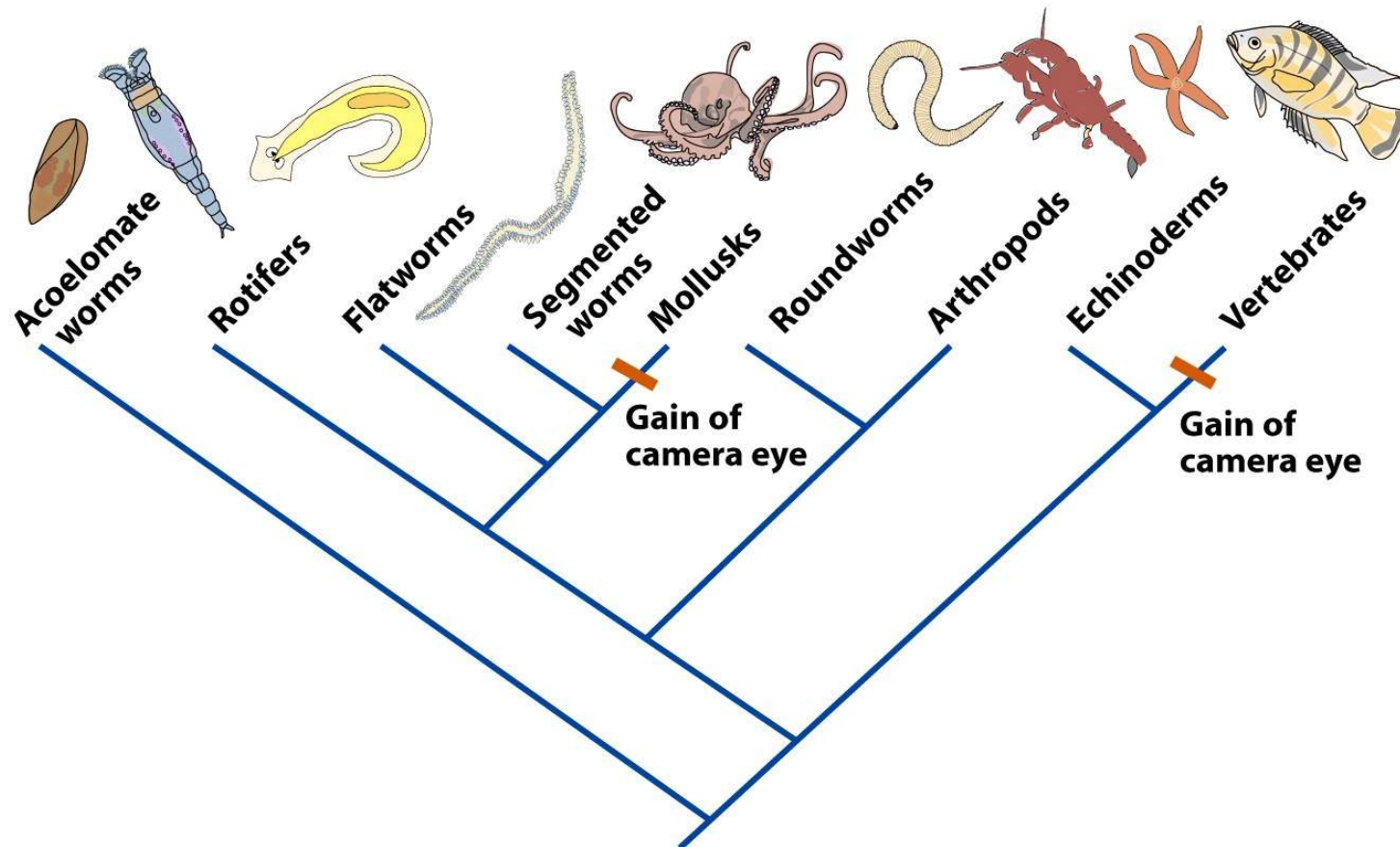
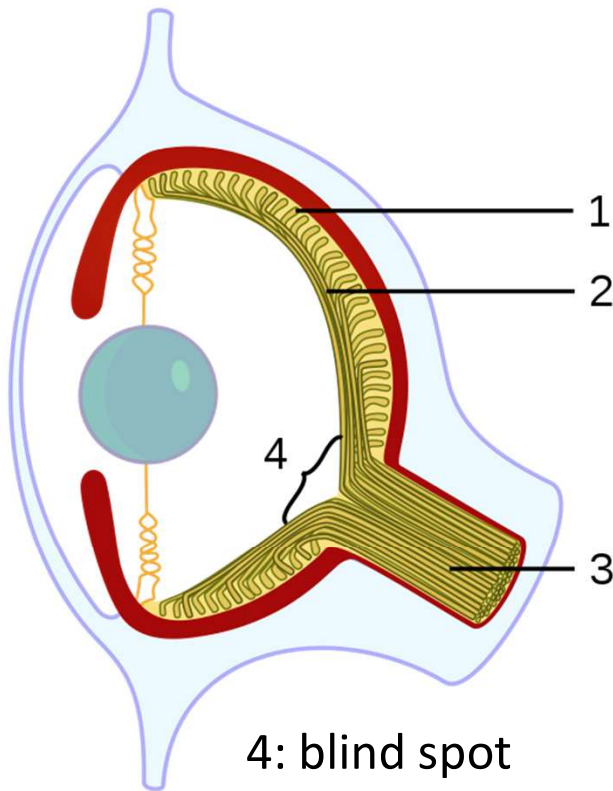


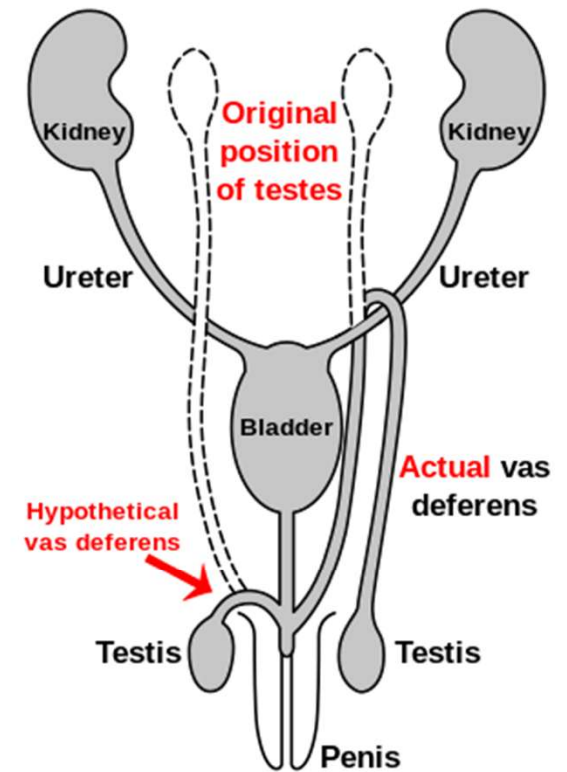
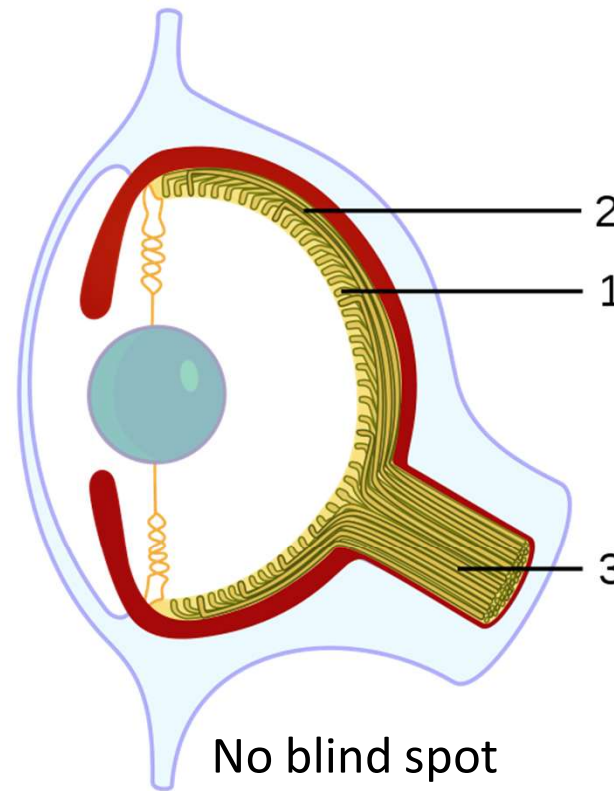
Figure 4-6b Evolutionary Analysis, 4/e
© 2007 Pearson Prentice Hall, Inc.

'Adaptive', yes, but 'optimal design'?

Vertebrate eye



Octopus eye



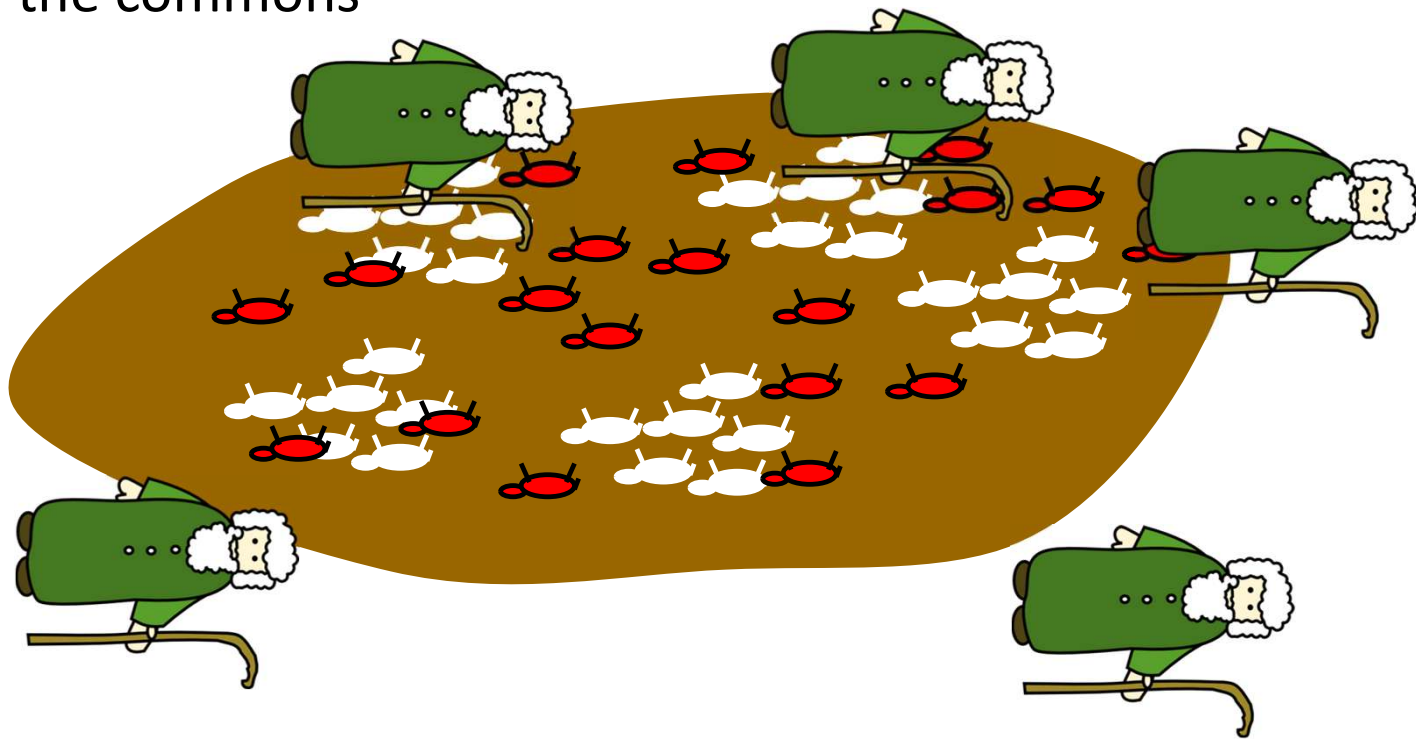
‘Adaptive’, yes, but ‘useful’?

Sexual selection can favour traits that reduce survival (chapter 11)



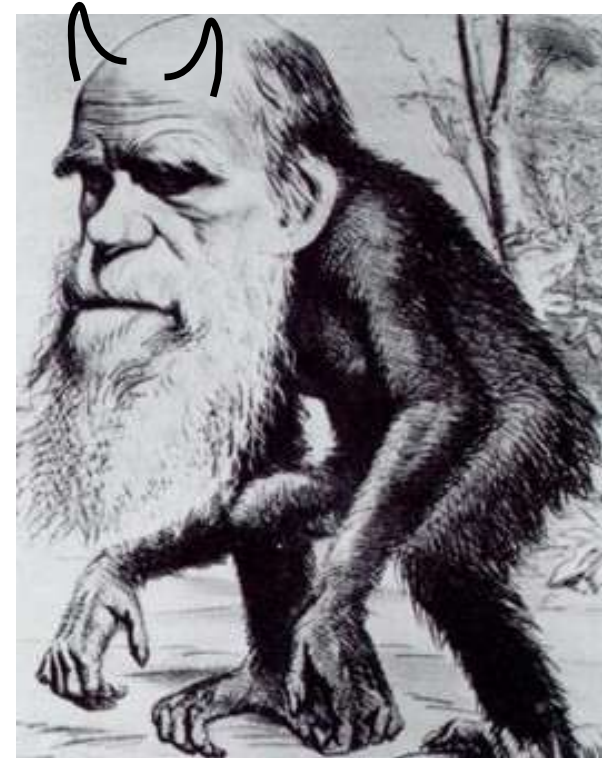
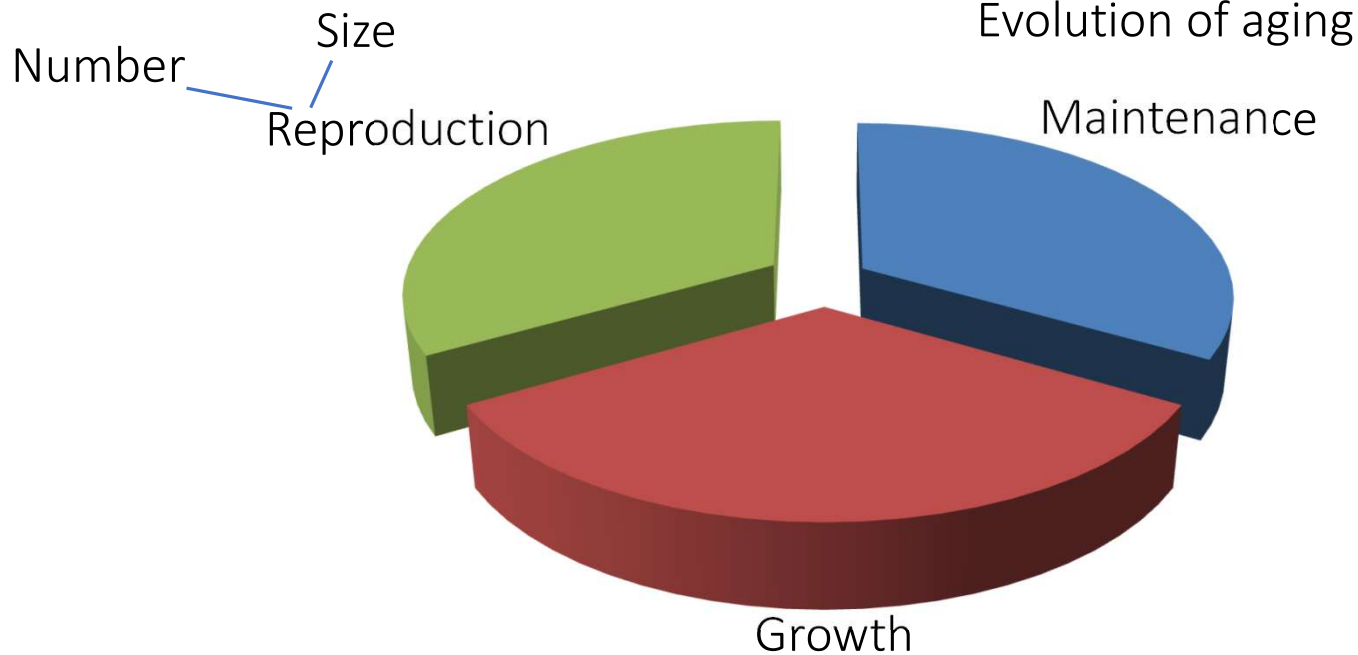
'Adaptive', yes, but 'for whom'?

Usually selection at the level of the individual, but bad for the group → tragedy of the commons



'Adaptive', yes, but 'for free'?

No! Trade-offs and constraints preclude
'Darwinian demon'



So adaptation, yes, but

Not always 'optimal design';

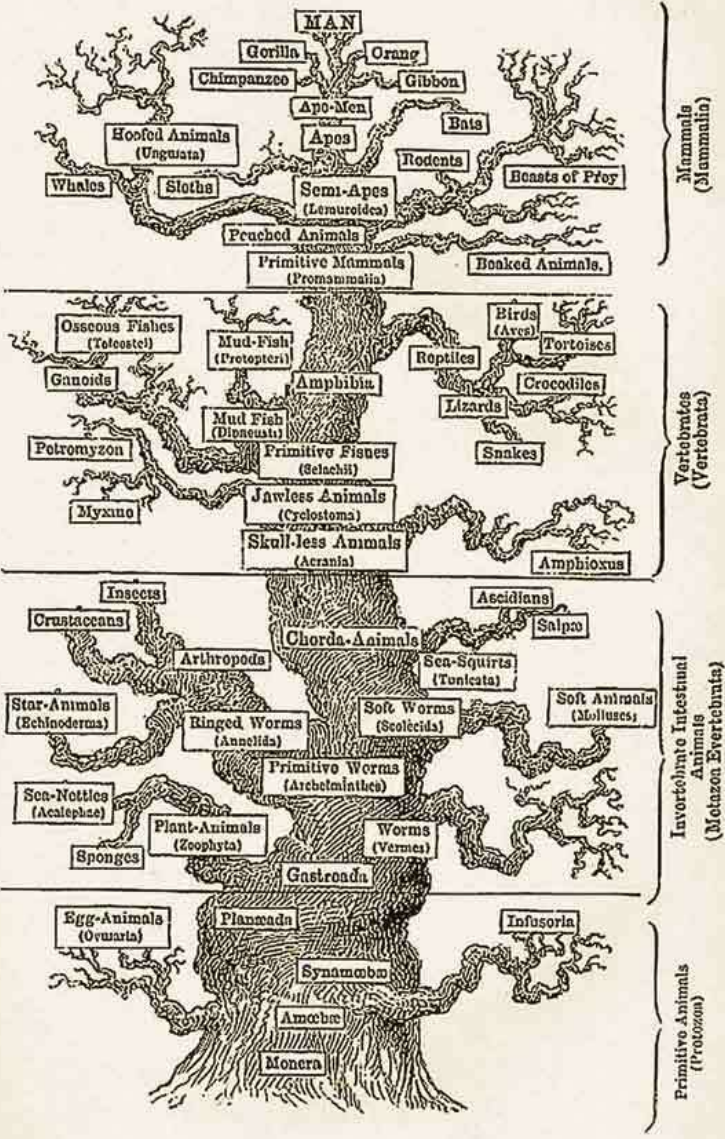
Not always 'useful';

Usually for the individual, not for the group;

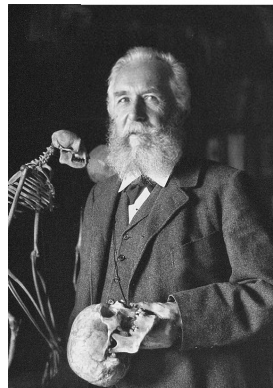
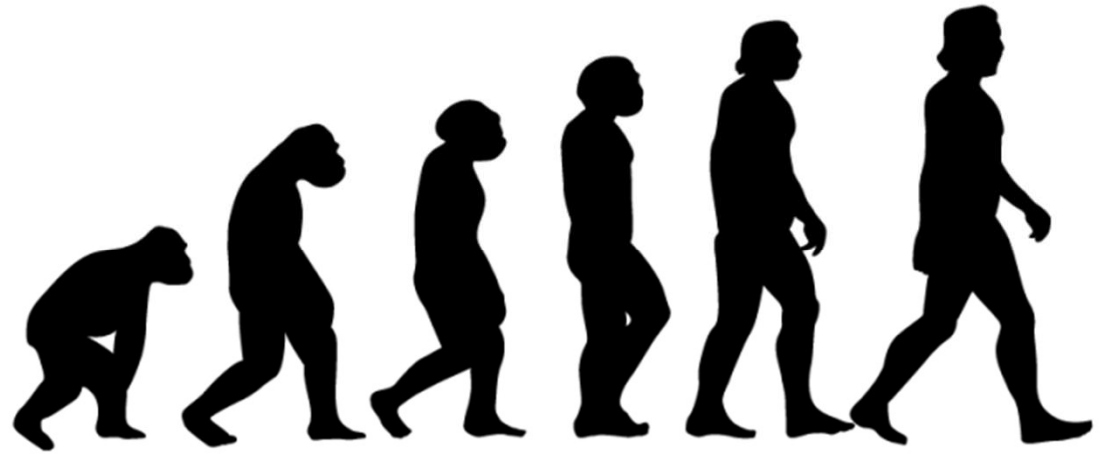
And not for free.

Does evolution nevertheless lead to progress?

PEDIGREE OF MAN.

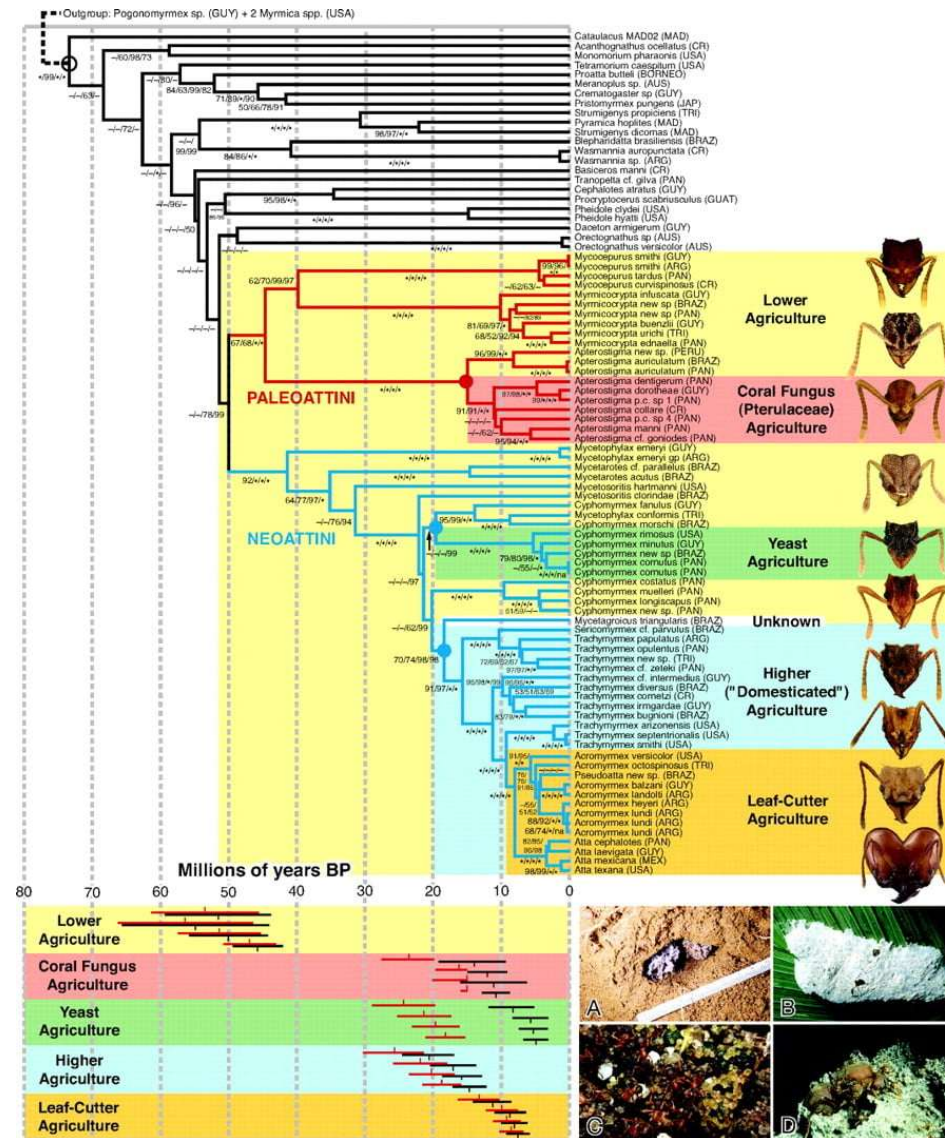


“Ladders of progress”



Ernst Haeckel

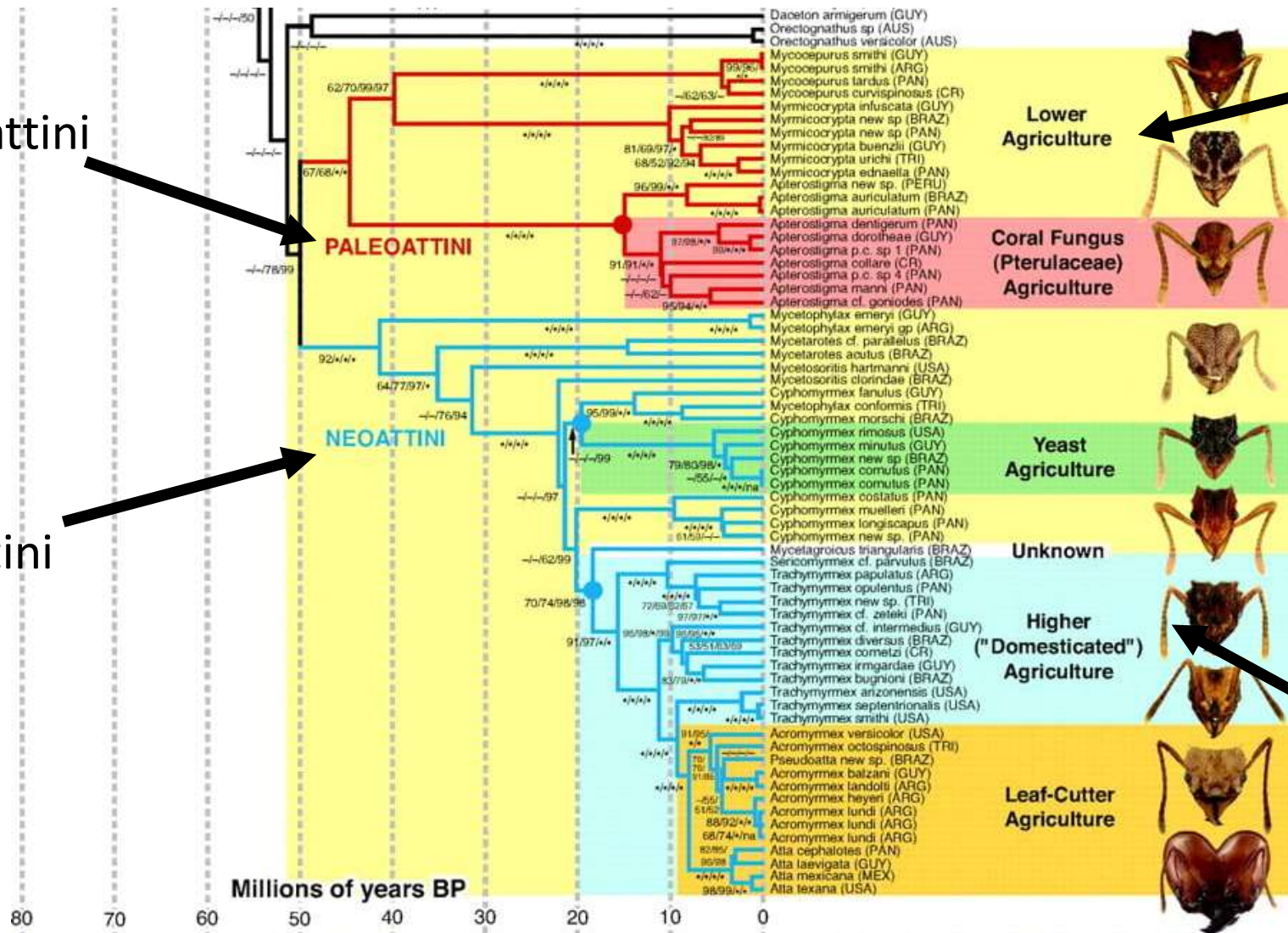
Phylogeny of fungus-growing ants



Brady and Schultz, 2008

Paleoattini

Neoattini



Lower agriculture

Lower Agriculture

Coral Fungus (Pterulaceae) Agriculture

Yeast Agriculture

Unknown

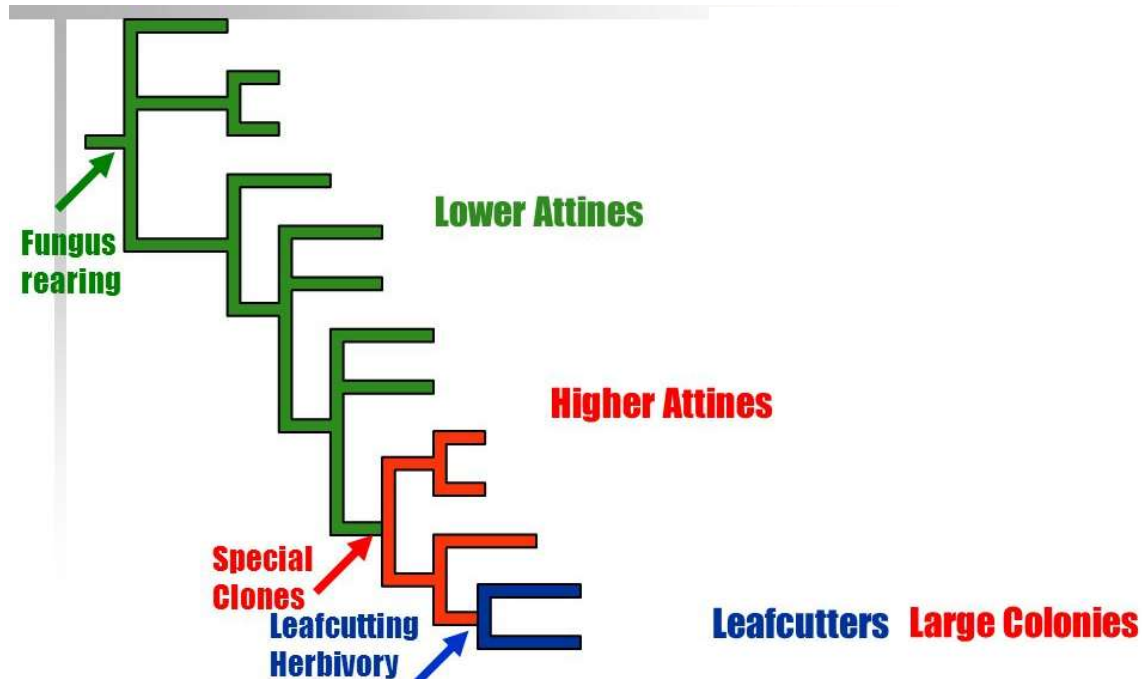
Higher ("Domesticated") Agriculture

Leaf-Cutter Agriculture

Higher agriculture

Brady and Schultz, 2008

“Ladders of progress”



Conspicuous synapomorphies are used to describe a group as ‘advanced’, ‘higher’ or ‘derived’.

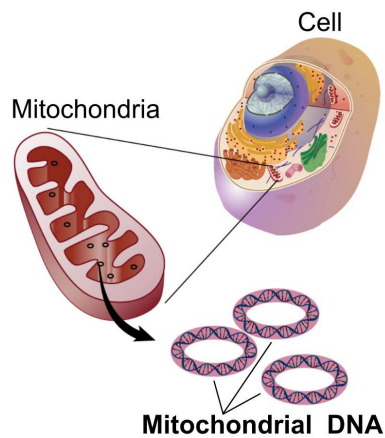
Conversely, taxa sharing plesiomorphic conspicuous character state are used to describe a group as ‘primitive’, ‘lower’ or ‘ancestral’.

This is wrong!

All extant taxa have a mixture of derived and ancestral character states.

Progress and evolution

The idea that evolution is associated with progress is still widespread. And we can see that complexity has increased in the course of evolution... For example, during the Major Transitions in Evolution.



Is evolution more than science?

- The history of Darwinism
- What is evolutionary biology and what is it not?
- What makes evolution difficult to accept for some people?

Is evolution more than science?

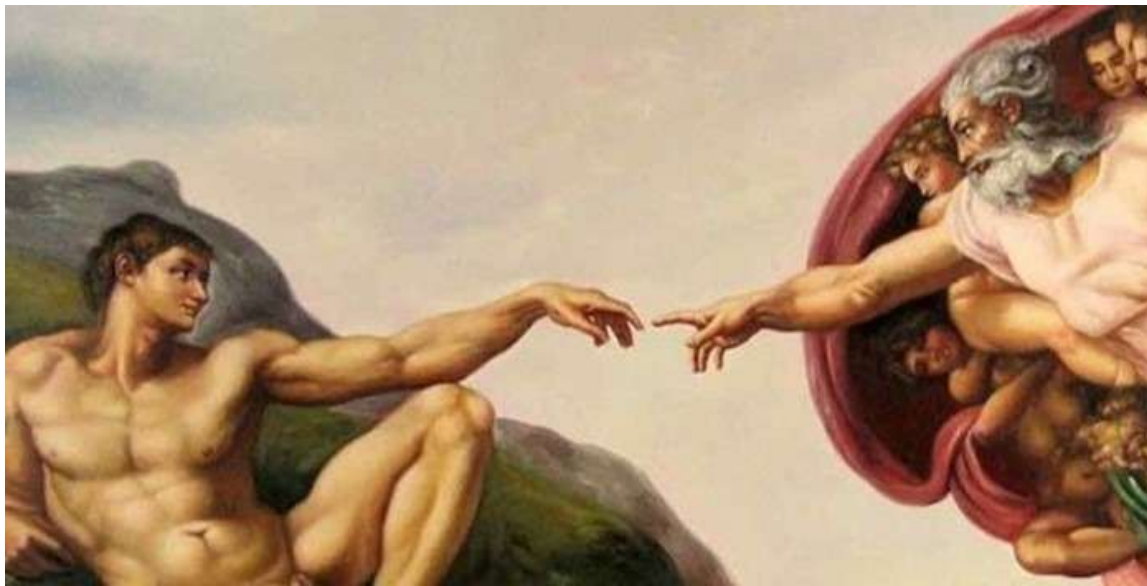
- The history of Darwinism
- What is evolutionary biology and what is it not?
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The theory of evolution is not compatible with a literal interpretation of Creation



Moreover, evolution was also about the origin and evolution of man

Questions about the origin of man traditionally belonged to the domain of religion and other world views.



The theory of evolution gave answers to why questions!

- Why do plants grow towards light?
- Why are males often aggressive?
- Why are females choosy?

Why questions traditionally belonged to the domain of religion/world view → the theory of evolution as a world view?

Why questions (ultimate questions): adaptation as a result of natural selection

- Goal orientedness in classification of Tinbergen (1963) → mechanism of natural selection
- ‘To catch more light’: ‘natural selection favoured plants with the characteristic *phototropism*, because those plants could produce more offspring’.

	Non-Historical	Historical
Proximate	Mechanisms (Causation)	Development (Ontogeny)
Ultimate	Survival Value (Function)	Evolution (Phylogeny)

Therefore, ultimate questions in biology are 'how' questions

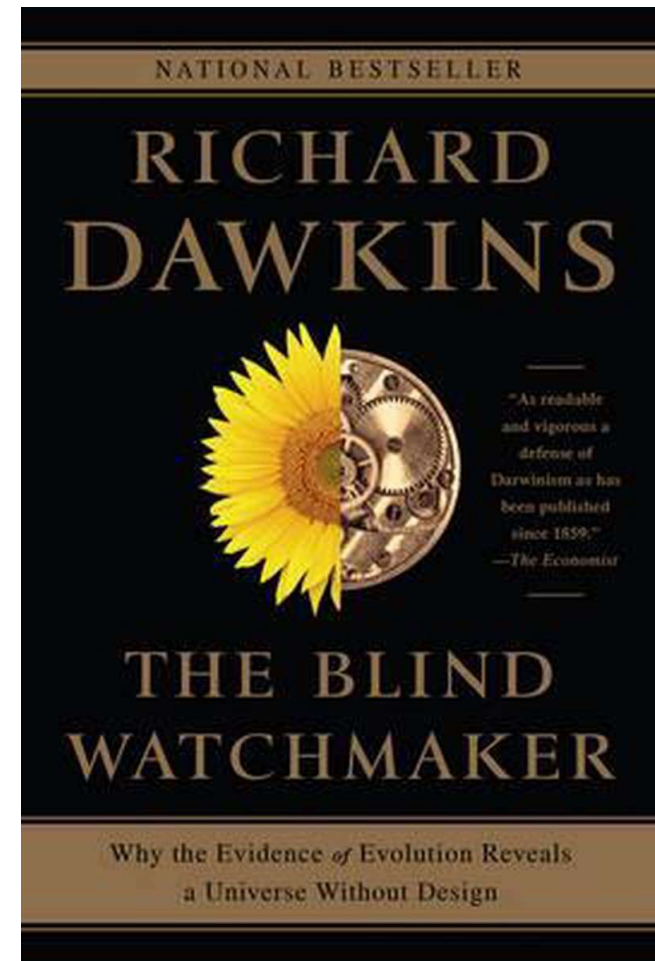
- How can we explain functionality of morphology and behaviour as a way to increase survival and reproduction?
- 'Ultimate explanation' $\neq$ 'definitive explanation'
- You can always ask the question if there is a deeper ('more ultimate') why question, and that question itself is one which science cannot answer.

Methodological naturalism

- **‘Rule of the game’** of science: only natural explanations are valid!
- So no appeal on supernatural explanations

Methodological naturalism works two ways

- No goal orientedness in the mechanism of natural selection
- Dawkins' statement goes beyond: no mechanism for design → no design in the universe.
- Not testable with scientific method, because the absence of indications for design is no absolute proof for the absence of design.



Evolutionary biology is also about human behaviour

- Morality
- Cultural evolution
- Distinction between “tendency to” and “concrete elaboration of”?



Biologically determined (?)

Culturally determined (?)

Role of free will (?)

The naturalistic fallacy

What do we expect from evolutionary biology?

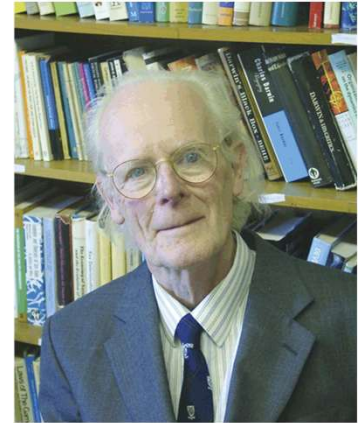
In the past, evolutionary explanations have been used as a moral justification

Social Darwinism: *survival of the fittest* to justify imperialism, racism, eugenics and social inequality

But there are still misunderstandings on the implications of evolution

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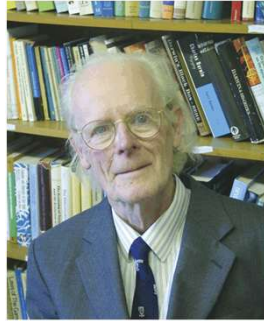
Science, Ideology and Myth



Recently, after giving a broadcast on Charles Darwin, I received through the post a pamphlet, 'Why are there "Gays" at all? Why hasn't evolution eliminated "Gayness" millions of years ago?', by Don Smith. The pamphlet points to a

John Maynard Smith '*Did Darwin get it right?*' 1988

Maynard Smith rigorously rejects this idea:

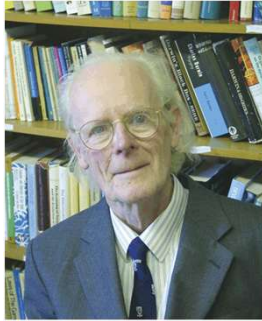


“If people have despised gays because gayness does not contribute to biological fitness, they have been wrong to do so.

It would be as sensible to persecute mathematicians because an ability to solve differential equations does not contribute to fitness.

A scientific theory – Darwinism or any other – has nothing to say about the value of a human being.”

Science and myth



Keep stories with a moral message (myths) and science separated.

The origin of species is not a book with moral guidelines for our behaviour.

Likewise, the Bible is not a book with scientific explanations for the natural world.

Naturalistic fallacy: “what is natural is good”

(and a variant: Moralistic fallacy: “what is good must be natural”)

Evolution is the grand story of biology

- At its start, Darwinism was a mixture between science and ideology
- After the rediscovery of Mendel's laws, evolution got a solid basis, and became a true science.
- Basis for historical and functional explanations of organismal traits
- Theory of natural selection offers ultimate explanations for adaptation and agency of organisms → fundamental distinction with other sciences
- Tinbergen's 'ultimate explanation' no 'definitive explanation' without further ado, because one can always ask 'more ultimate' questions that are outside the domain of science.